

Jerrold E. Marsden and Anthony J. Tromba

# **Vector Calculus**

## **Fifth Edition**

### **Chapter 5:**

## **Double and Triple Integrals**

### 5.1 Introduction

## 5.1 Introduction

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### Key Points in this Section.

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1. If  $R = [a, b] \times [c, d]$  is a rectangle in the plane and  $f : R \rightarrow \mathbb{R}$  is a non-negative function, then

$$\iint_R f(x, y) dA = \iint_R f(x, y) dx dy$$

is the volume of the region under the graph of  $f$  and above the rectangle  $R$ . This is an ‘informal’ definition in that it assumes one knows about volumes. A ‘rigorous’ definition is given in §5.2.

**2. Cavalieri's Principle.** Suppose that one is given the following data (see Figure 5.1.1):

- (a) A solid  $S$ ,
- (b) An  $x$ -axis in space,
- (c) Planes  $P_x$  perpendicular to the  $x$ -axis cutting  $S$  in regions  $R_x$  with areas  $A(x)$  for  $x$  ranging between  $x = a$  and  $x = b$ .
- (d) Then the volume of  $S$  is

$$V = \int_a^b A(x) dx.$$

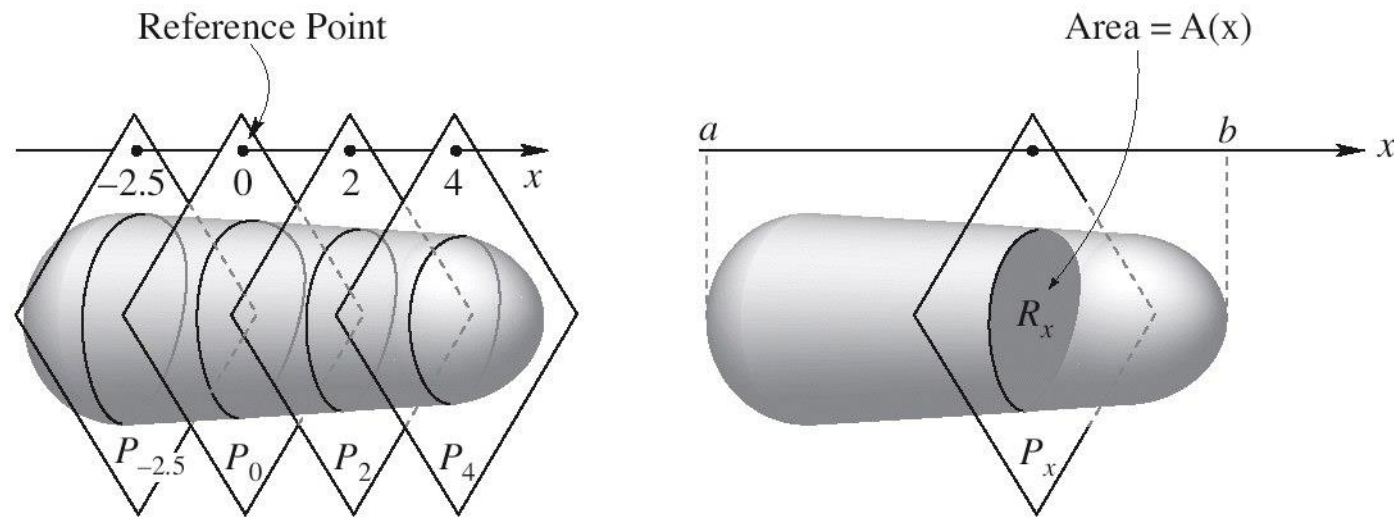
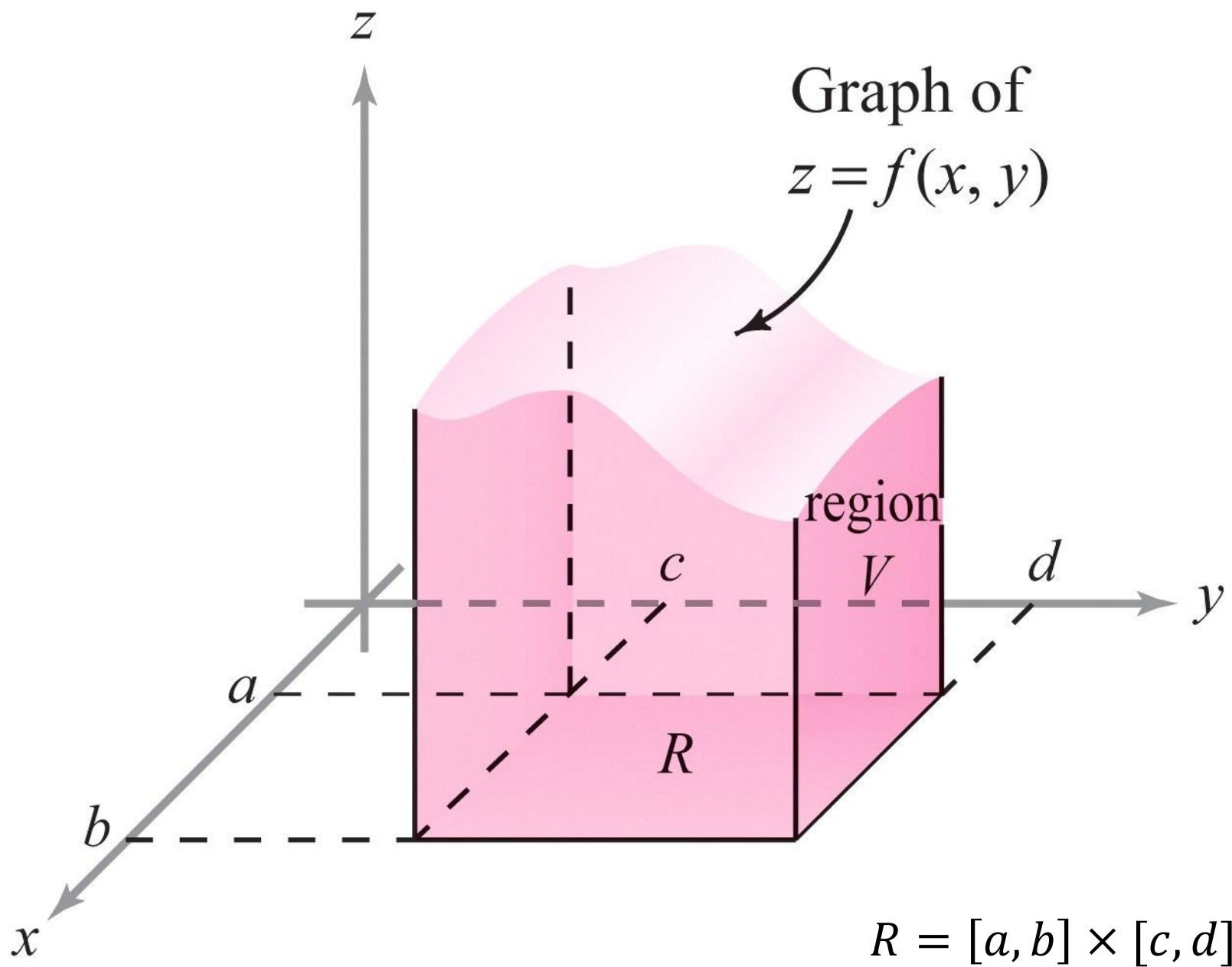


FIGURE 5.1.1. The data used in Cavalieri's principle:  $\text{Volume} = \int_a^b A(x) dx$

3. **Iterated Integrals.** Using slices along the  $x$  and  $y$ -axes, together with the interpretation of the one-variable integral as an area, Cavalieri's principle leads to the double integral written as iterated integrals:

$$\iint_R f(x, y) dA = \int_a^b \left[ \int_c^d f(x, y) dy \right] dx = \int_c^d \left[ \int_a^b f(x, y) dx \right] dy.$$



**Double Integrals** The volume of the region above  $R$  and under the graph of a nonnegative function  $f$  is called the *(double) integral* of  $f$  over  $R$  and is denoted by

$$\iint_R f(x, y) dA, \quad \text{or} \quad \iint_R f(x, y) dx dy.$$

$z$

$$R = [0,1] \times [0,1]$$

$$f(x, y) = 1 - x$$

$$\iint_R f(x, y) \, dA = \frac{1}{2}$$

$$z = 1 - x$$

$R$

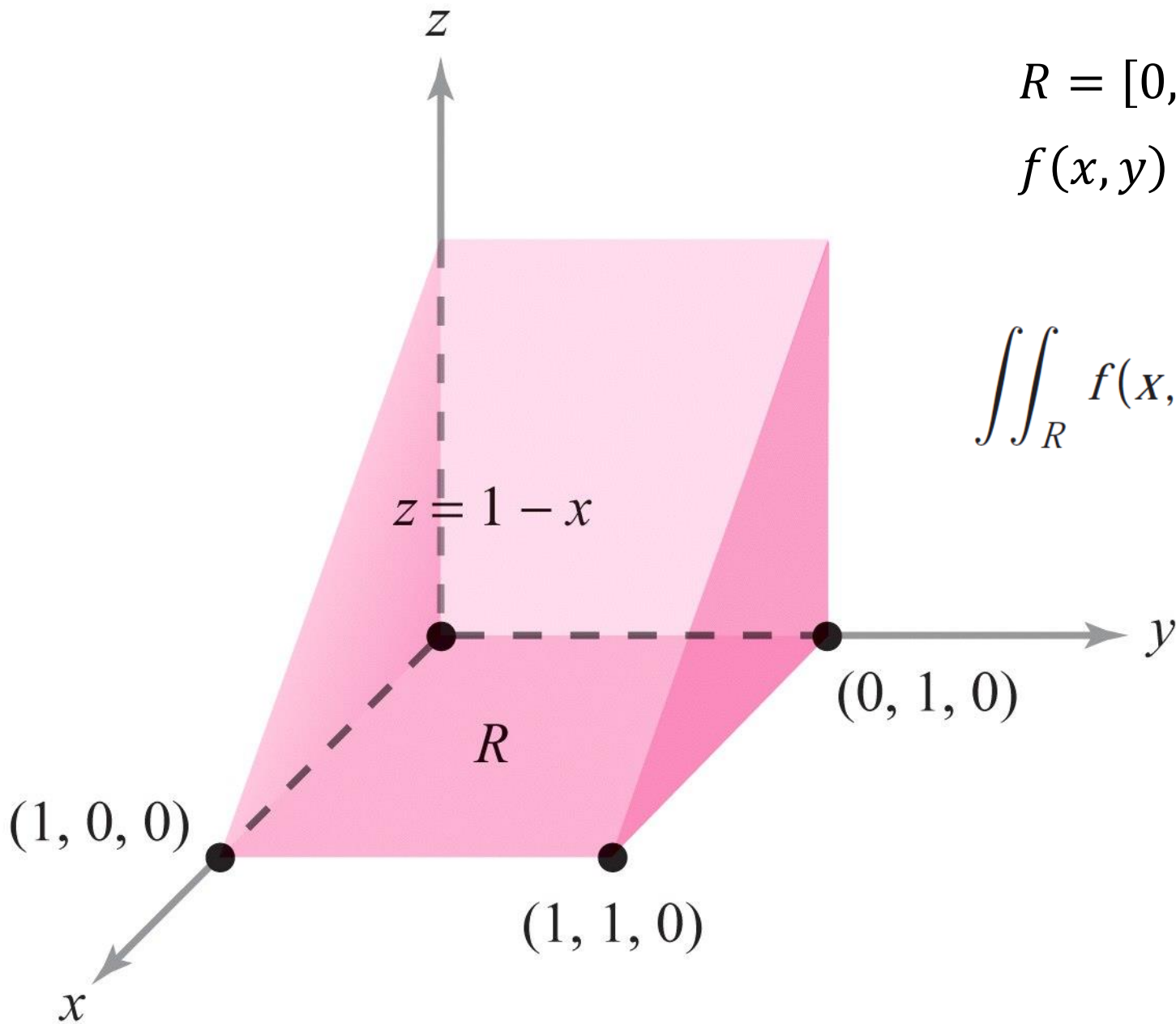
$(1, 0, 0)$

$(1, 1, 0)$

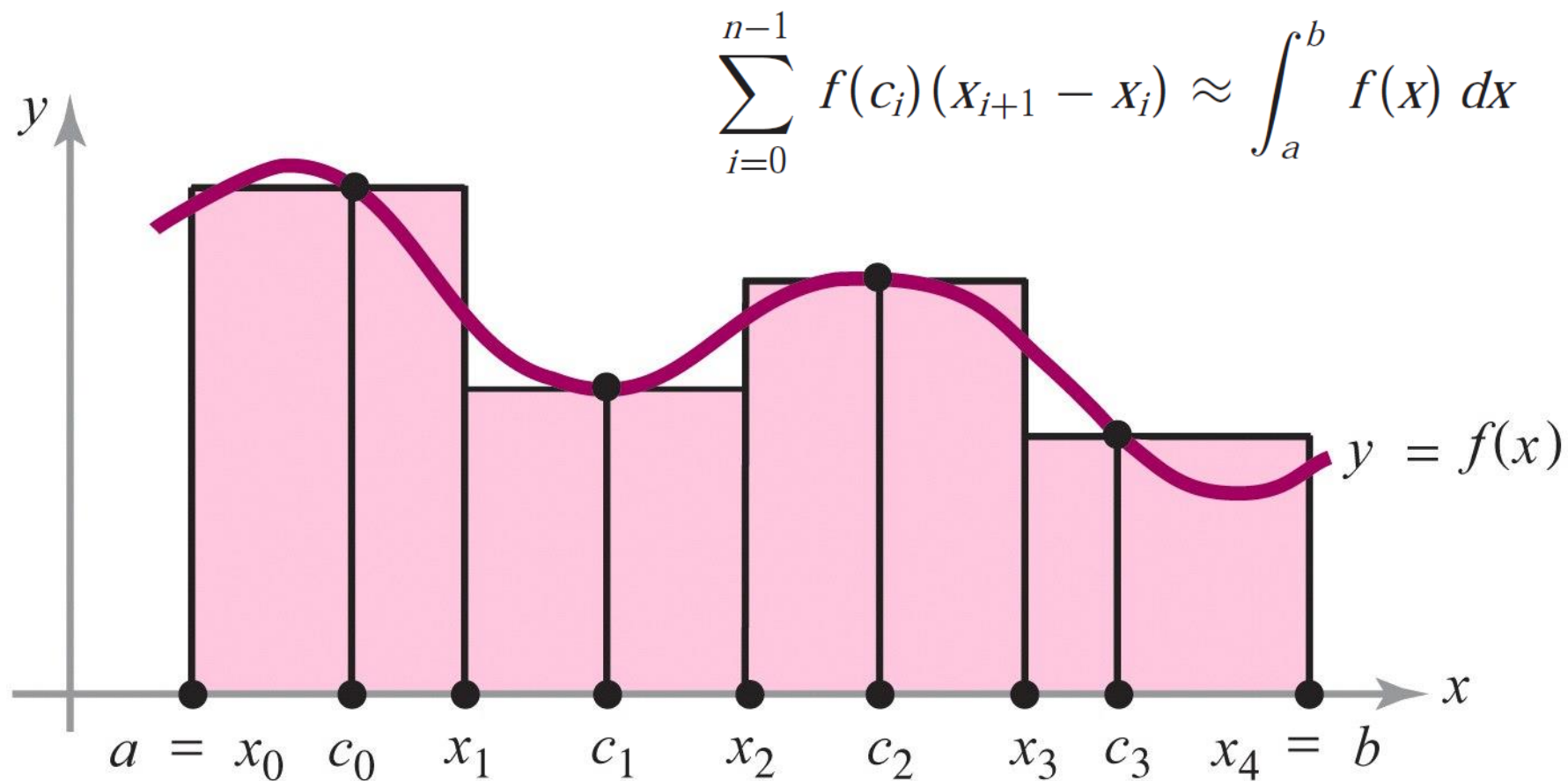
$(0, 1, 0)$

$x$

$y$



# Integral as limit of Riemann sums

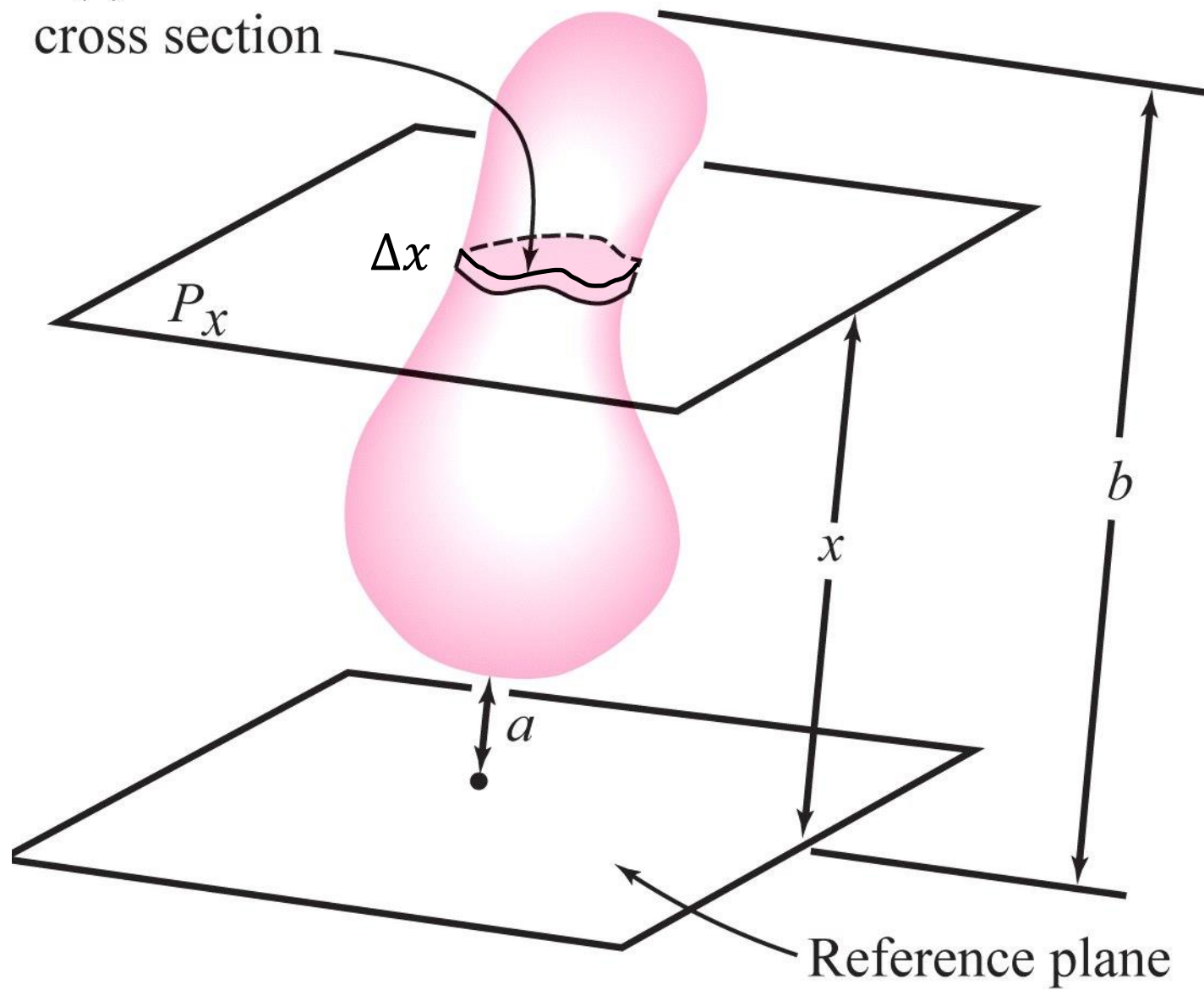


$$a = x_0 < x_1 < \cdots < x_n = b$$

$$c_i \in [x_i, x_{i+1}]$$



$A(x)$  = area of  
cross section

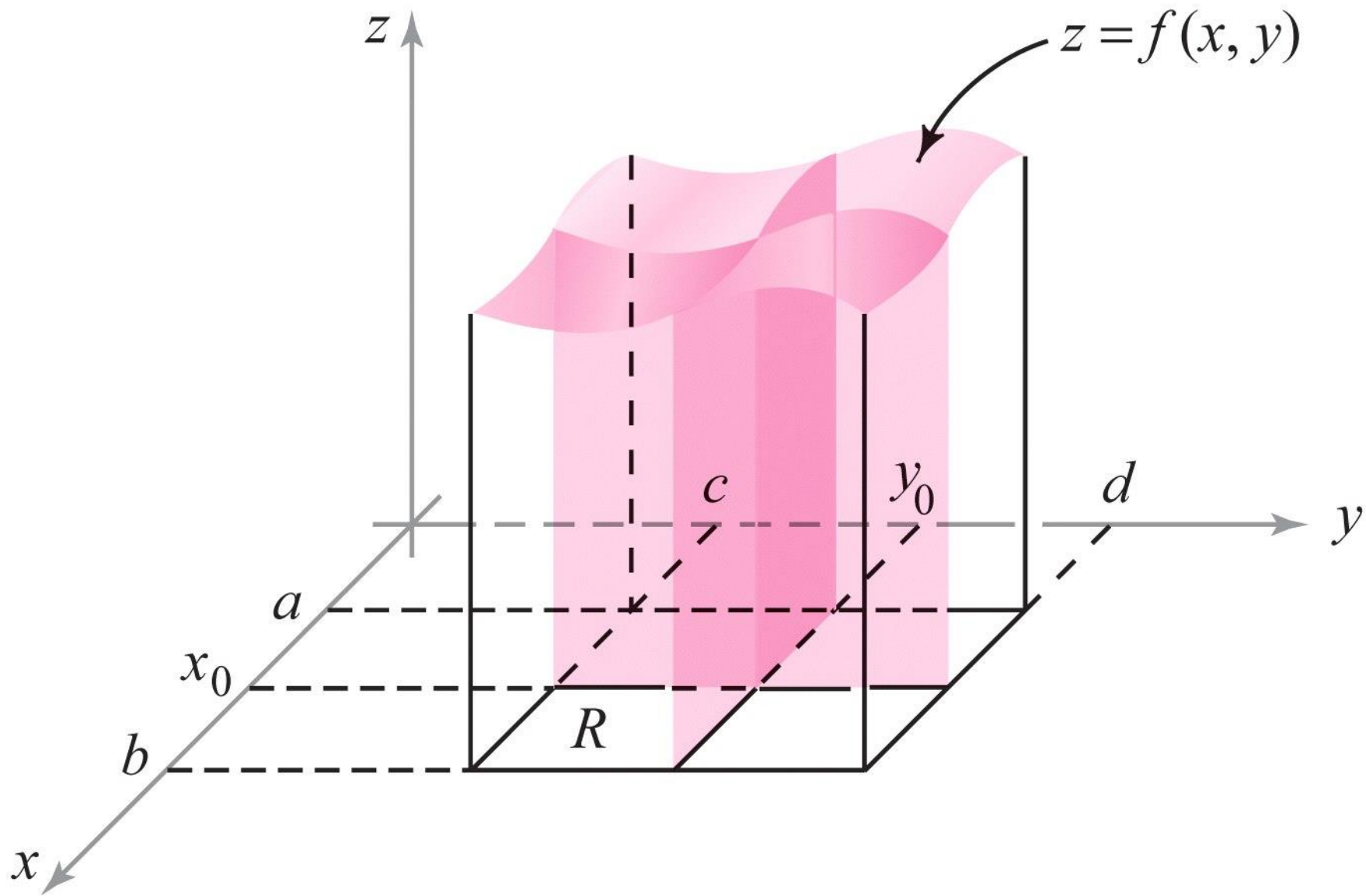


**The Slice Method — Cavalieri's Principle** Let  $S$  be a solid and, for  $x$  satisfying  $a \leq x \leq b$ , let  $P_x$  be a family of parallel planes such that:

1.  $S$  lies between  $P_a$  and  $P_b$ ;
2. The area of the slice of  $S$  cut by  $P_x$  is  $A(x)$ .

Then the volume of  $S$  is equal to

$$\int_a^b A(x) dx.$$



## Double and Iterated Integrals

Iterated integral by integrating first with respect to  $y$  and then integrating the result with respect to  $x$

$$\iint_R f(x, y) dA = \int_a^b \left[ \int_c^d f(x, y) dy \right] dx$$

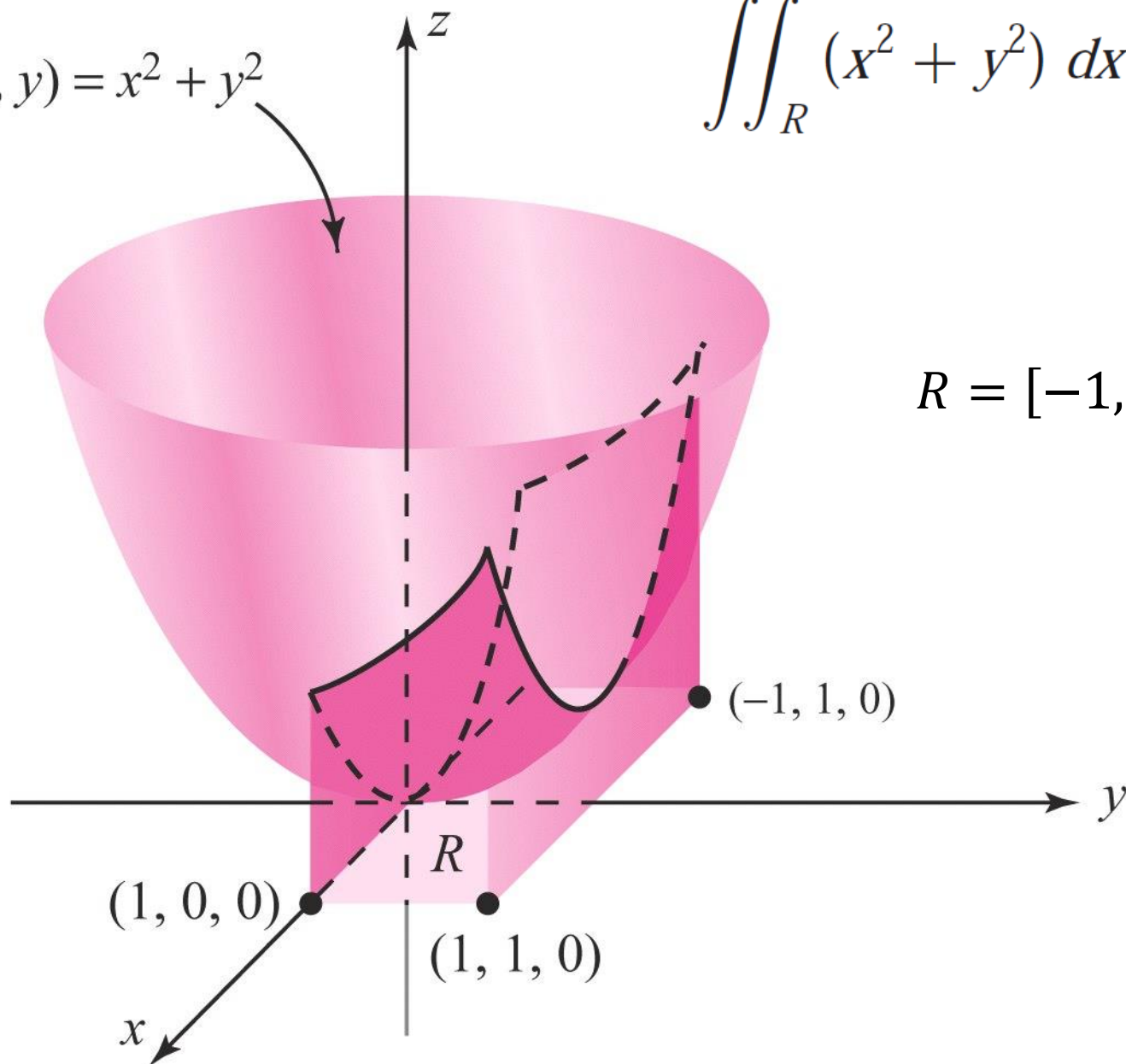
Iterated integral by integrating first with respect to  $x$  and then integrating the result with respect to  $y$

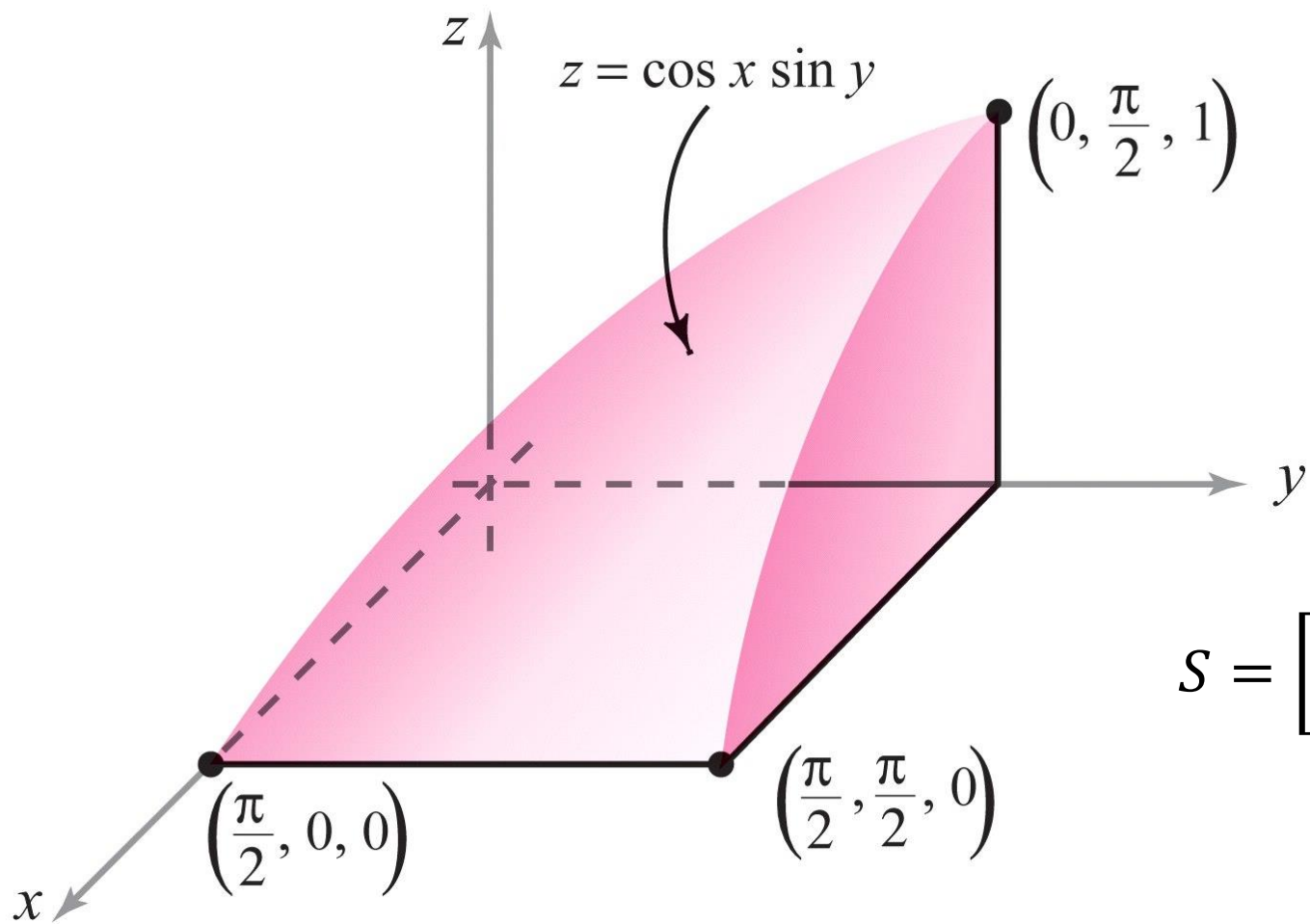
$$\iint_R f(x, y) dA = \int_c^d \left[ \int_a^b f(x, y) dx \right] dy$$

$$z = f(x, y) = x^2 + y^2$$

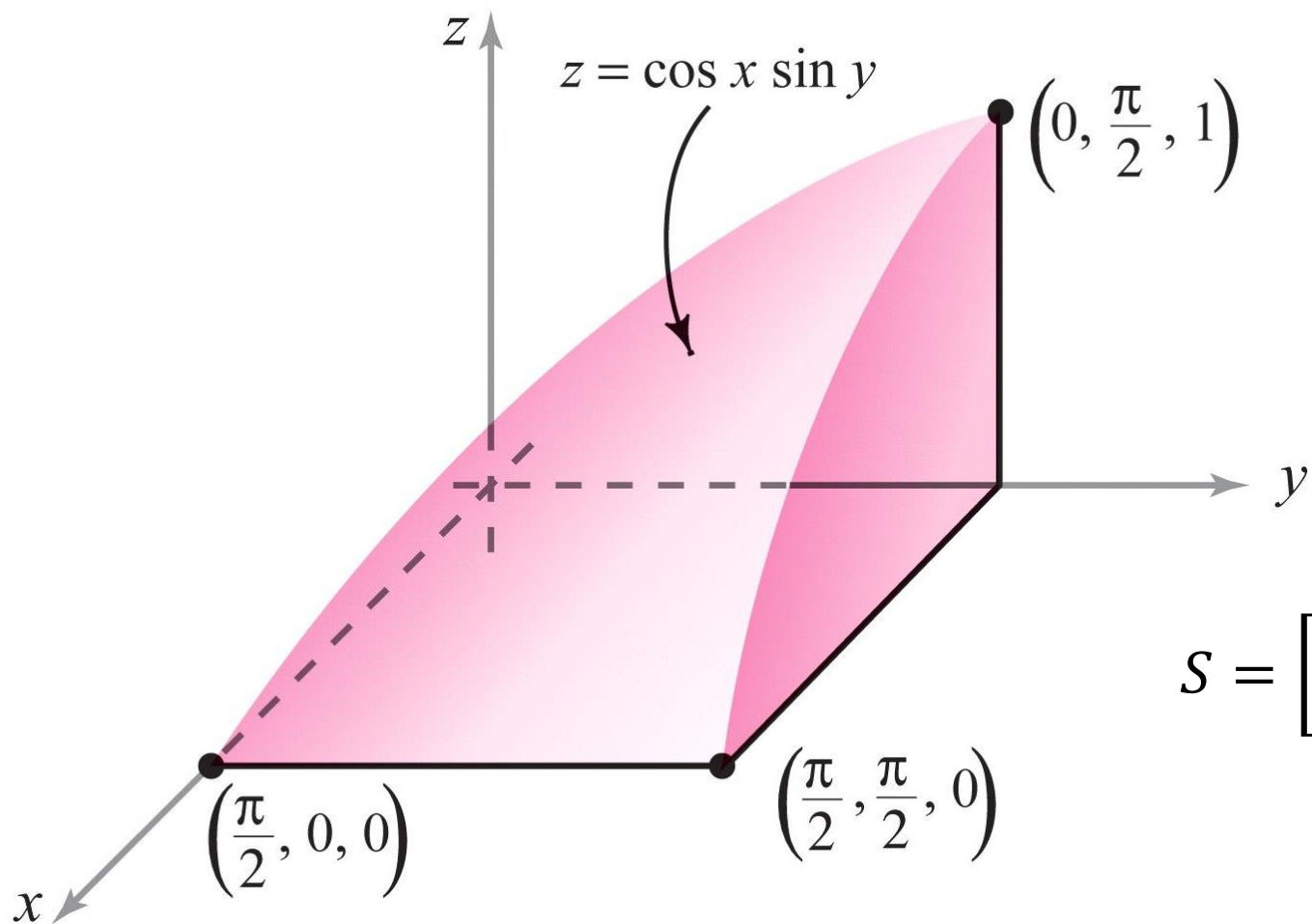
$$\iint_R (x^2 + y^2) \, dx \, dy = \frac{4}{3}$$

$$R = [-1, 1] \times [0, 1]$$



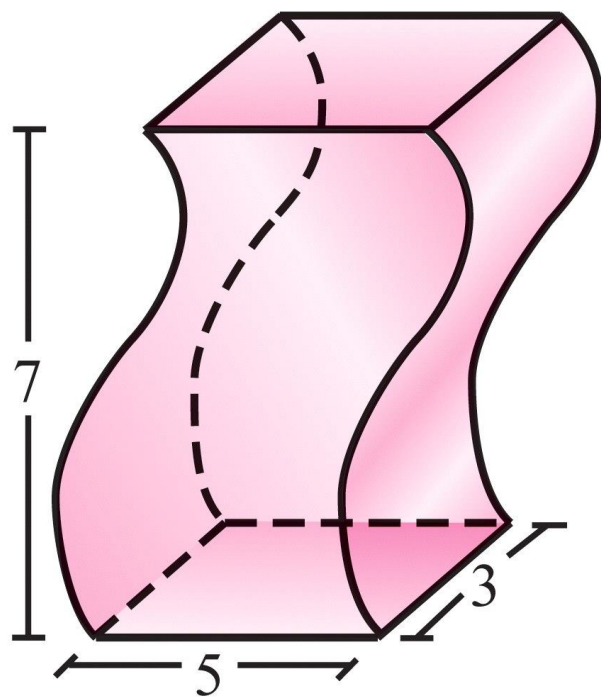
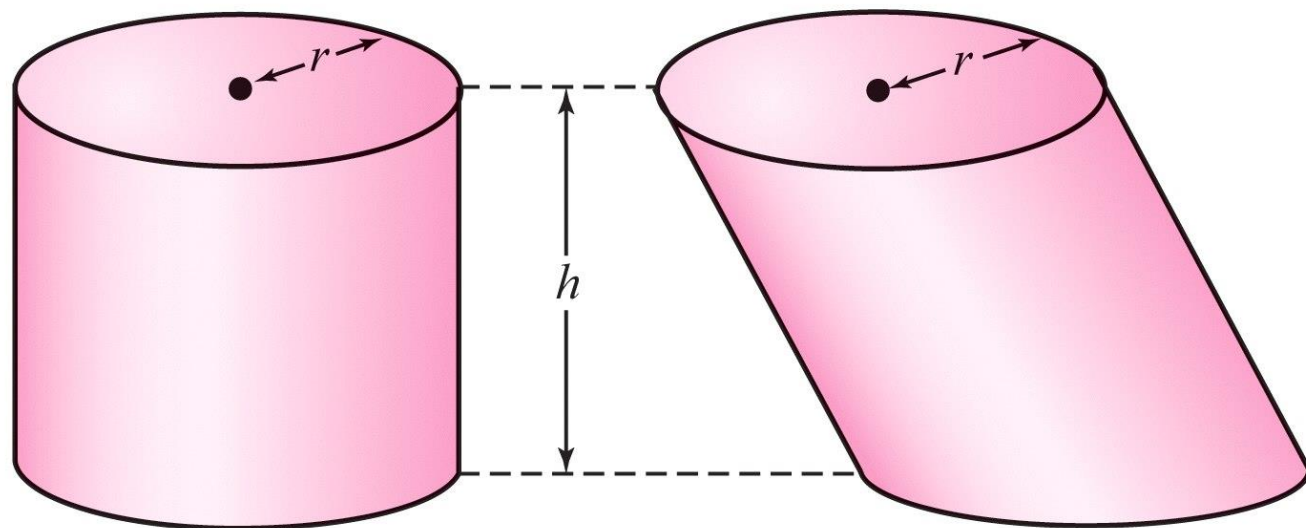


$$S = \left[0, \frac{\pi}{2}\right] \times \left[0, \frac{\pi}{2}\right]$$



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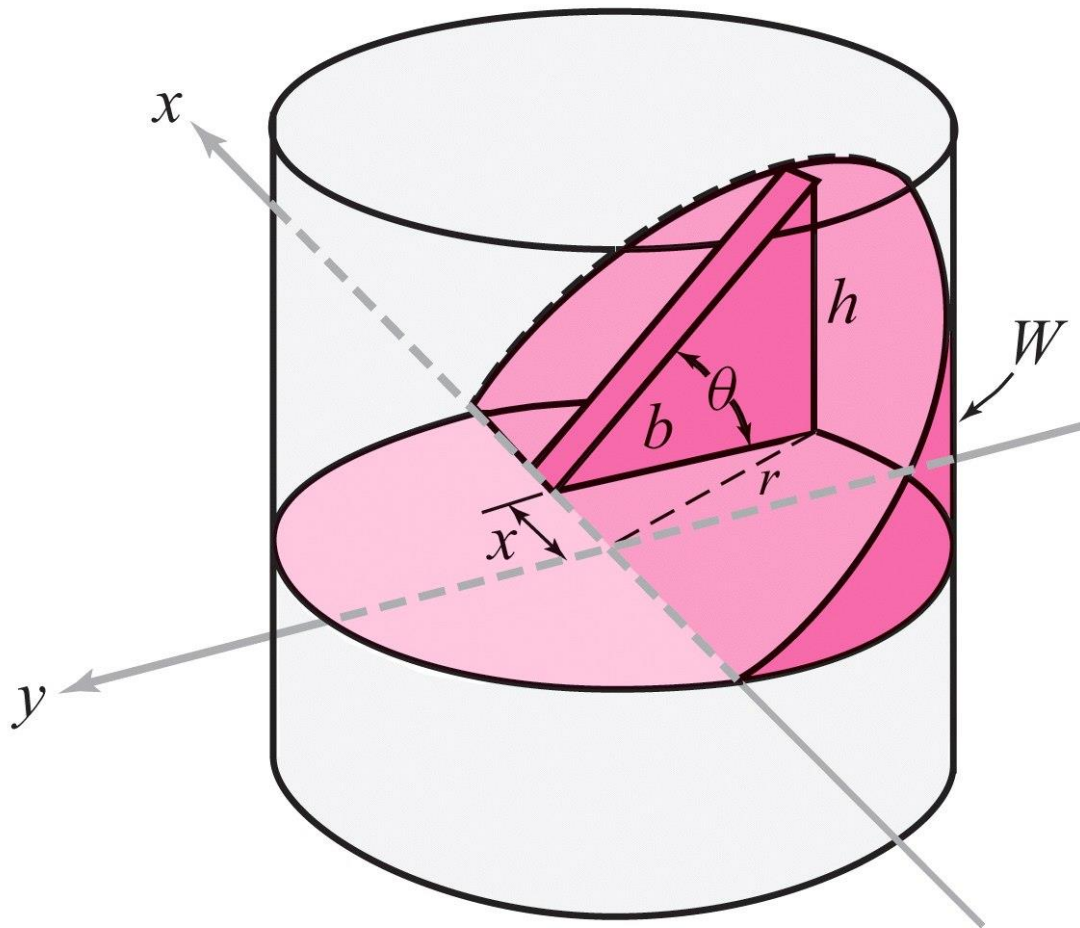
$$\begin{aligned} \iint_S \cos x \sin y \, dx \, dy &= \int_0^{\pi/2} \left[ \int_0^{\pi/2} \cos x \sin y \, dx \right] dy \\ &= \int_0^{\pi/2} \sin y \left[ \int_0^{\pi/2} \cos x \, dx \right] dy = \int_0^{\pi/2} \sin y \, dy = 1 \end{aligned}$$



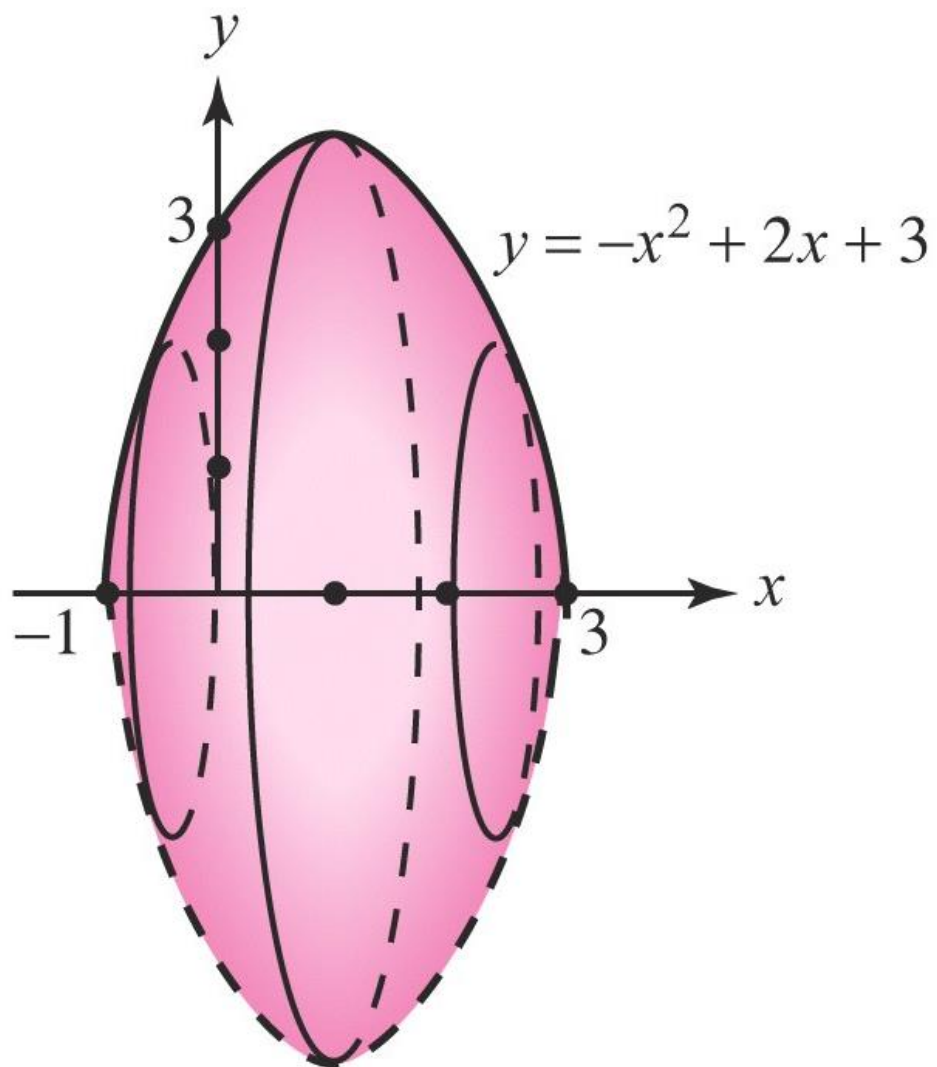


## Exercise

A lumberjack cuts out a wedge-shaped piece  $W$  of a cylindrical tree of radius  $r$  obtained by making two saw cuts to the tree's center, one horizontally and one at an angle  $\theta$ . Compute the volume of the wedge  $W$  using Cavalieri's principle.



Solution:  $\frac{2}{3} r^3 \tan \theta$



(b)

$$V = \frac{512\pi}{15}$$

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# **Vector Calculus**

## **Fifth Edition**

### **Chapter 5:**

### **Double and Triple Integrals**

#### 5.2 The Double Integral Over a Rectangle

## 5.2 The Double Integral over a Rectangle

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### Key Points in this Section.

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1. A **Riemann sum** for a function  $f$  defined on a rectangle  $R = [a, b] \times [c, d]$  has the form

$$S_n = \sum_{j,k=0}^{n-1} f(\mathbf{c}_{jk}) \Delta x \Delta y,$$

where  $R$  is divided into  $n^2$  equal sub-rectangles obtained by dividing  $[a, b]$  and  $[c, d]$  into  $n$  equal parts, and where  $\mathbf{c}_{jk}$  is a point chosen in the  $jk^{\text{th}}$  sub-rectangle,  $0 \leq j, k \leq n - 1$ , of width  $\Delta x$  and height  $\Delta y$ .

2. **Definition of the Integral.** If  $\lim_{n \rightarrow \infty} S_n = S$  exists and is independent of the choice of  $\mathbf{c}_{jk}$ ,  $f$  is called **integrable** over  $R$  and the limit is denoted

$$\iint_R f(x, y) dA, \text{ or } \int \int_R f(x, y) dx dy, \text{ or } \int \int_R f dA.$$

3. Continuous functions as well as functions that are bounded and that are continuous except along a finite union of graphs of functions (of either  $x$  or  $y$ ) are integrable.
4. The integral is linear in its argument and is additive with respect to the region. It also satisfies

$$\left| \iint_R f dA \right| \leq \iint_R |f| dA$$

5. For  $f \geq 0$ , the rigorous definition in point 2 justifies interpreting  $\iint_R f dA$  as the volume of the region under the graph of  $f$  and over  $R$ , as well as giving a theoretical foundation for the definition of the volume of a region.
6. **Fubini's Theorem** states that for  $f$  continuous, the reduction to iterated integrals holds:

$$\iint_R f(x, y) dA = \int_a^b \left[ \int_c^d f(x, y) dy \right] dx = \int_c^d \left[ \int_a^b f(x, y) dx \right] dy.$$

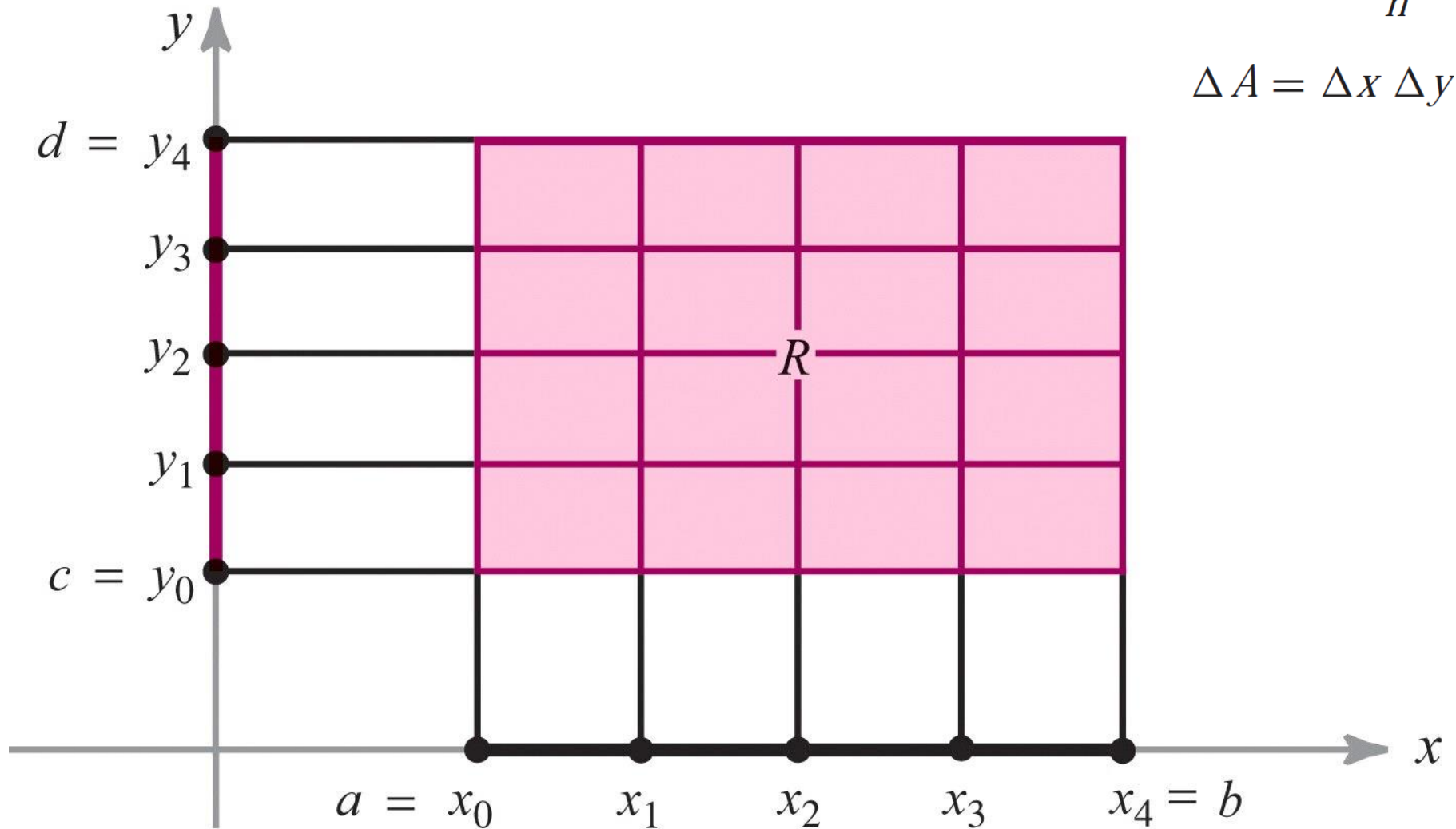
A similar result holds for bounded functions with discontinuities along a finite number of graphs provided the iterated integrals exist.

$$R = [a, b] \times [c, d]$$

$$\Delta x = x_{j+1} - x_j = \frac{b - a}{n}$$

$$\Delta y = y_{k+1} - y_k = \frac{d - c}{n}$$

$$\Delta A = \Delta x \Delta y$$



## Double integral as limit of Riemann sums

$$S_n = \sum_{j,k=0}^{n-1} f(\mathbf{c}_{jk}) \Delta x \Delta y \qquad \mathbf{c}_{jk} \in [x_j, x_{j+1}] \times [y_k, y_{k+1}]$$

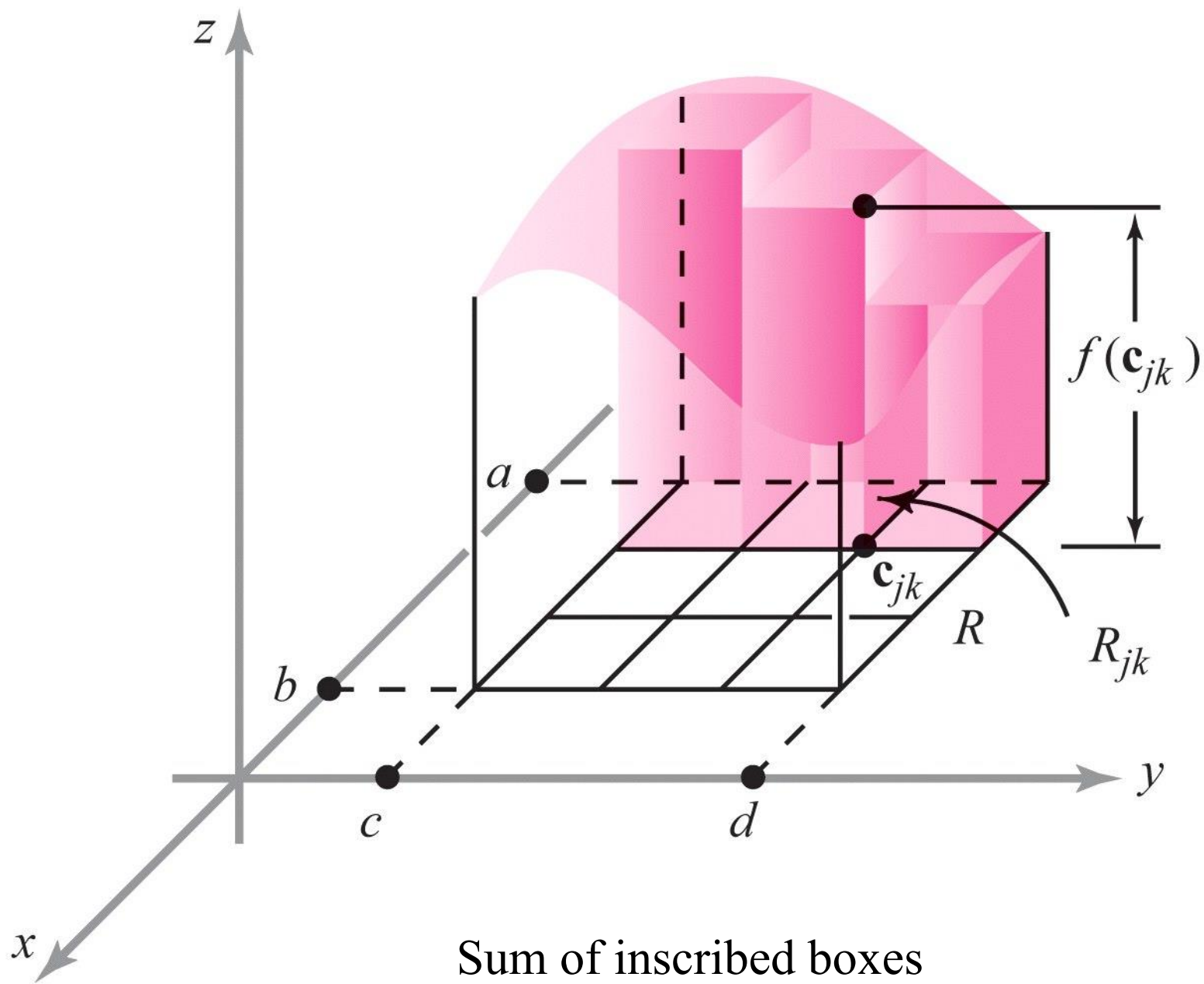
**DEFINITION: Double Integral** If the sequence  $\{S_n\}$  converges to a limit  $S$  as  $n \rightarrow \infty$  and if the limit  $S$  is the same for any choice of points  $\mathbf{c}_{jk}$  in the rectangles  $R_{jk}$ , then we say that  $f$  is *integrable* over  $R$  and we write

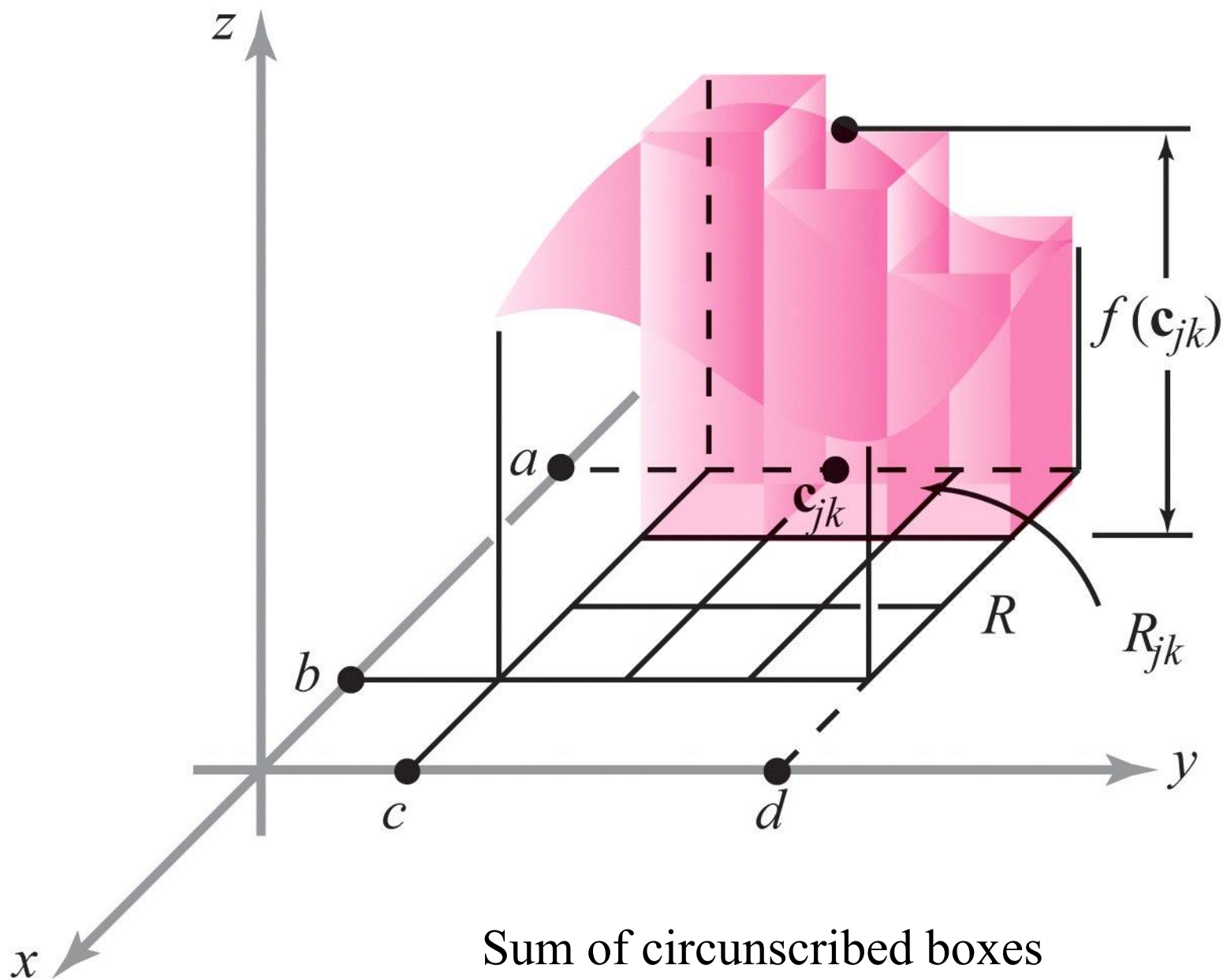
$$\iint_R f(x, y) dA, \quad \iint_R f(x, y) dx dy, \quad \text{or} \quad \iint_R f dx dy$$

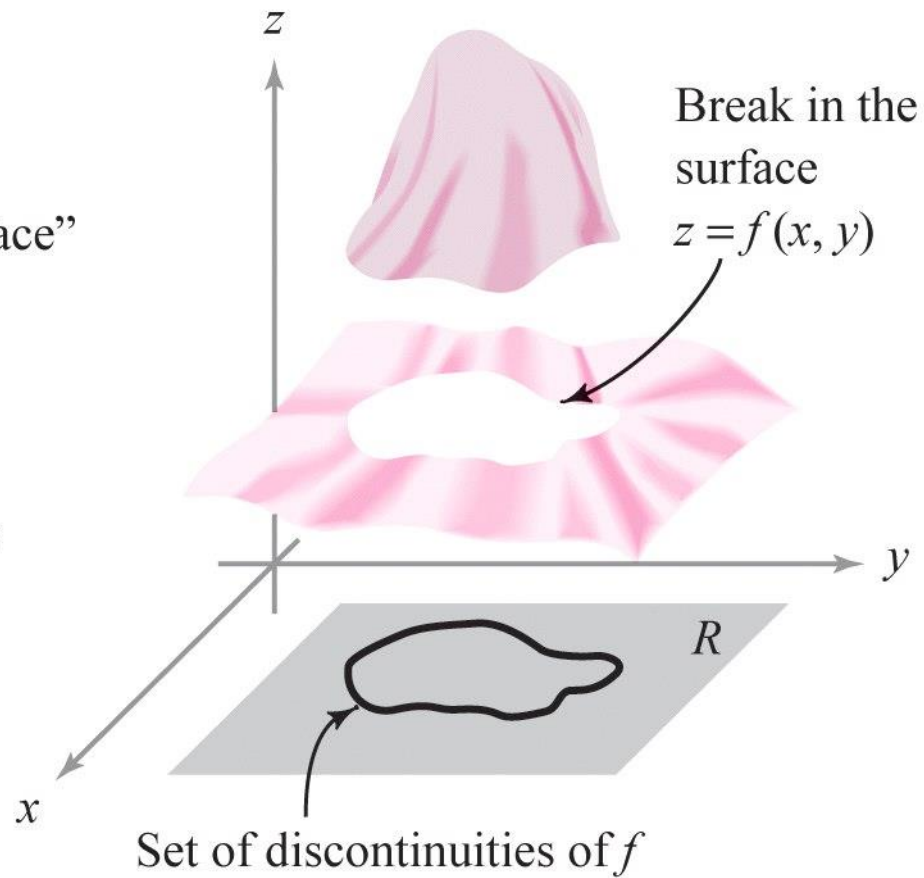
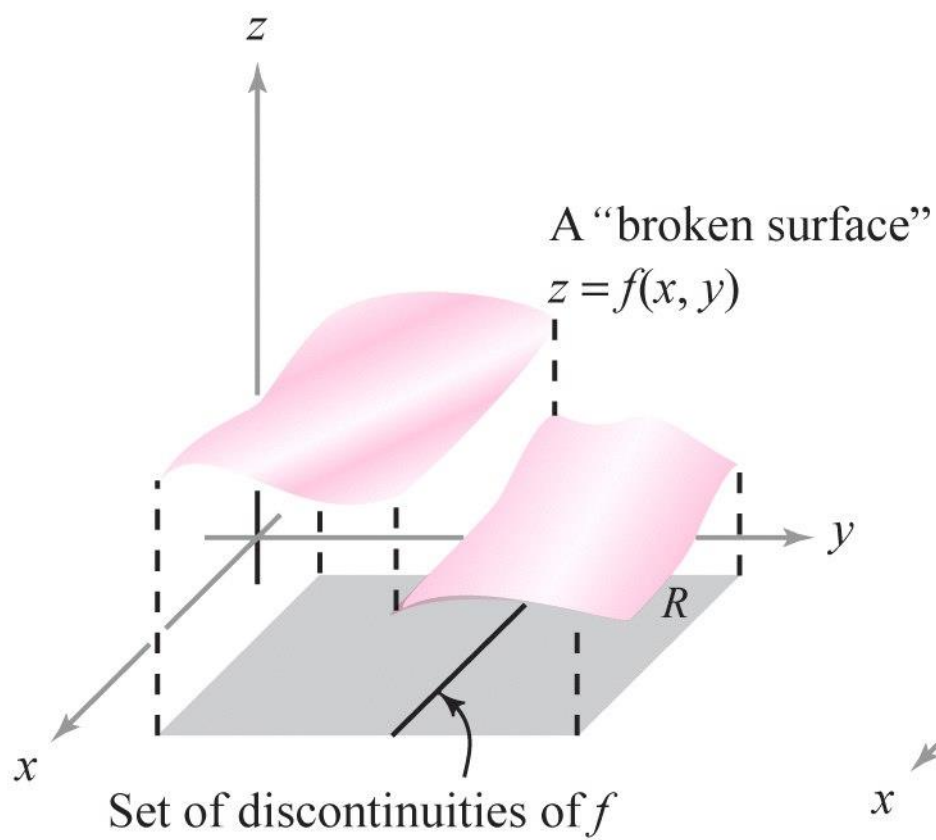
for the limit  $S$ .

**THEOREM 1** Any continuous function defined on a closed rectangle  $R$  is integrable.









**THEOREM 2: Integrability of Bounded Functions** Let  $f: R \rightarrow \mathbb{R}$  be a bounded real-valued function on the rectangle  $R$ , and suppose that the set of points where  $f$  is discontinuous lies on a finite union of graphs of continuous functions. Then  $f$  is integrable over  $R$ .

# Properties

## (i) Linearity

$$\iint_R [f(x, y) + g(x, y)] dA = \iint_R f(x, y) dA + \iint_R g(x, y) dA.$$

## (ii) Homogeneity

$$\iint_R cf(x, y) dA = c \iint_R f(x, y) dA.$$

## (iii) Monotonicity If $f(x, y) \geq g(x, y)$ , then

$$\iint_R f(x, y) dA \geq \iint_R g(x, y) dA.$$

## (iv) Additivity If $R_i, i = 1, \dots, m$ , are pairwise disjoint rectangles such that $f$ is bounded and integrable over each $R_i$ and if $Q = R_1 \cup R_2 \cup \dots \cup R_m$ is a rectangle, then $f: Q \rightarrow \mathbb{R}$ is integrable over $Q$ and

$$\iint_Q f(x, y) dA = \sum_{i=1}^m \iint_{R_i} f(x, y) dA.$$

**THEOREM 3: Fubini's Theorem** Let  $f$  be a continuous function with a rectangular domain  $R = [a, b] \times [c, d]$ . Then

$$\int_a^b \int_c^d f(x, y) dy dx = \int_c^d \int_a^b f(x, y) dx dy = \iint_R f(x, y) dA. \quad (4)$$

**proof** We shall first show that

$$\int_a^b \int_c^d f(x, y) \, dy \, dx = \iint_R f(x, y) \, dA.$$

Let  $c = y_0 < y_1 < \cdots < y_n = d$  be a partition of  $[c, d]$  into  $n$  equal parts. Define

$$F(x) = \int_c^d f(x, y) \, dy.$$

Then

$$F(x) = \sum_{k=0}^{n-1} \int_{y_k}^{y_{k+1}} f(x, y) \, dy.$$

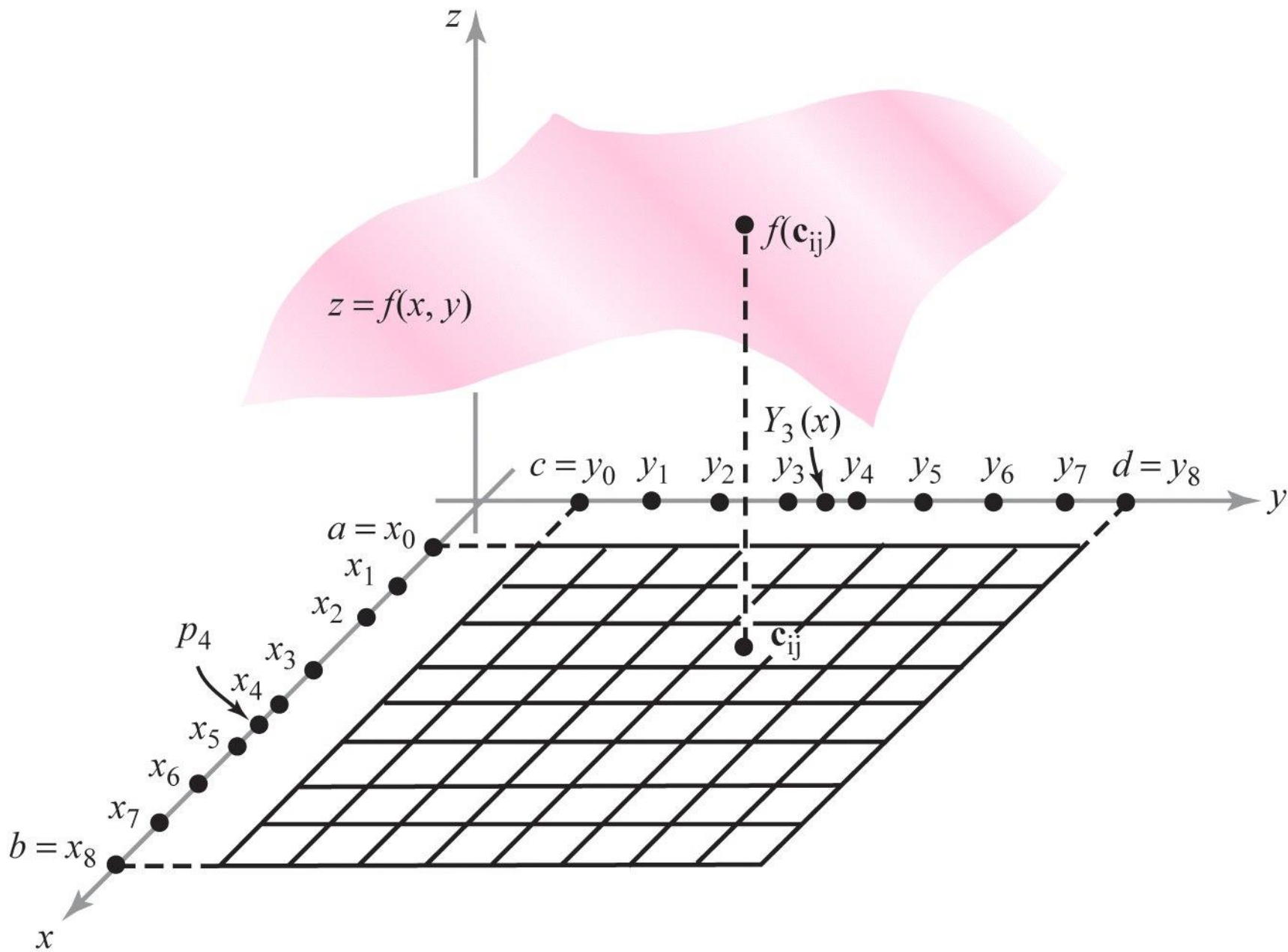
Using the integral version of the mean-value theorem,<sup>3</sup> for each fixed  $x$  and for each  $k$  we have

$$\int_{y_k}^{y_{k+1}} f(x, y) \, dy = f(x, Y_k(x)) (y_{k+1} - y_k)$$

where the point  $Y_k(x)$  belongs to  $[y_k, y_{k+1}]$  and may depend on  $x$ ,  $k$ , and  $n$ .

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<sup>3</sup>This states that if  $g(x)$  is continuous on  $[a, b]$ , then  $\int_a^b g(x) \, dx = g(c) (b - a)$  for some point  $c \in [a, b]$ .





We have thus shown that

$$F(x) = \sum_{k=0}^{n-1} f(x, Y_k(x)) (y_{k+1} - y_k). \quad (5)$$

By the definition of the integral in one variable as a limit of Riemann sums,

$$\int_a^b F(x) dx = \int_a^b \left[ \int_c^d f(x, y) dy \right] dx = \lim_{n \rightarrow \infty} \sum_{j=0}^{n-1} F(p_j) (x_{j+1} - x_j),$$

where  $a = x_0 < x_1 < \dots < x_n = b$  is a partition of the interval  $[a, b]$  into  $n$  equal parts and  $p_j$  is any point in  $[x_j, x_{j+1}]$ . Setting  $\mathbf{c}_{jk} = (p_j, Y_k(p_j)) \in R_{jk}$ , we have [substituting  $p_j$  for  $x$  in equation (5)]

$$F(p_j) = \sum_{k=0}^{n-1} f(\mathbf{c}_{jk}) (y_{k+1} - y_k).$$

Therefore,

$$\begin{aligned}\int_a^b \int_c^d f(x, y) \, dy \, dx &= \int_a^b F(x) \, dx = \lim_{n \rightarrow \infty} \sum_{j=0}^{n-1} F(p_j) (x_{j+1} - x_j) \\ &= \lim_{n \rightarrow \infty} \sum_{j=0}^{n-1} \sum_{k=0}^{n-1} f(\mathbf{c}_{jk}) (y_{k+1} - y_k) (x_{j+1} - x_j) \\ &= \iint_R f(x, y) \, dA.\end{aligned}$$

Thus, we have proved that

$$\int_a^b \int_c^d f(x, y) \, dy \, dx = \iint_R f(x, y) \, dA.$$

By the same reasoning we can show that

$$\int_c^d \int_a^b f(x, y) \, dx \, dy = \iint_R f(x, y) \, dA.$$

These two conclusions are exactly what we wanted to prove. ■

## Exercises

Compute  $\iint_R (x^2 + y) \, dA$ , where  $R$  is the square  $[0, 1] \times [0, 1]$ .

Solution:  $\frac{5}{6}$

Let  $R$  be the rectangle  $[-2, 1] \times [0, 1]$  and let  $f$  be defined by  $f(x, y) = y(x^3 - 12x)$ ;  $f(x, y)$  takes on both positive and negative values on  $R$ . Evaluate the integral  $\iint_R f(x, y) \, dx \, dy = \iint_R y(x^3 - 12x) \, dx \, dy$ .

Solution:  $\frac{57}{8}$

**THEOREM 3': Fubini's Theorem** Let  $f$  be a bounded function with domain a rectangle  $R = [a, b] \times [c, d]$ , and suppose the discontinuities of  $f$  lie on a finite union of graphs of continuous functions. If the integral  $\int_c^d f(x, y) dy$  exists for each  $x \in [a, b]$ , then

$$\int_a^b \left[ \int_c^d f(x, y) dy \right] dx$$

exists and

$$\int_a^b \int_c^d f(x, y) dy dx = \iint_R f(x, y) dA.$$

Similarly, if  $\int_a^b f(x, y) dx$  exists for each  $y \in [c, d]$ , then

$$\int_c^d \left[ \int_a^b f(x, y) dx \right] dy$$

exists and

$$\int_c^d \int_a^b f(x, y) dx dy = \iint_R f(x, y) dA.$$

Thus, if all these conditions hold simultaneously,

$$\int_a^b \int_c^d f(x, y) dy dx = \int_c^d \int_a^b f(x, y) dx dy = \iint_R f(x, y) dA.$$

## Counterexample

Compute

$$\iint_R f(x, y) dx dy$$

where

$$R = [0, 2] \times [0, 1]$$

and

$$f(x, y) = \begin{cases} \frac{xy(x^2 - y^2)}{(x^2 + y^2)^3}, & (x, y) \neq (0, 0) \\ 0, & (x, y) = (0, 0) \end{cases}$$

## Counterexample

**First approach** We integrate first in the  $y$ -direction. For any  $x$ , we define  $A(x) = \int_0^1 f(x, y) dy$ . For now, we assume that  $x \neq 0$ . We make the substitution  $u = x^2 + y^2$ ,  $du = 2y dy$ , use  $x^2 - y^2 = x^2 - (u - x^2) = 2x^2 - u$ , and compute

$$\begin{aligned} A(x) &= \int_{x^2}^{x^2+1} \frac{x(2x^2 - u) du}{2u^3} = \int_{x^2}^{x^2+1} \left( \frac{x^3}{u^3} - \frac{x}{2u^2} \right) du = \left[ -\frac{x^3}{2u^2} + \frac{x}{2u} \right]_{u=x^2}^{u=x^2+1} \\ &= -\frac{x^3}{2(x^2+1)^2} + \frac{x}{2(x^2+1)} + \frac{1}{2x} - \frac{1}{2x} = \frac{x}{2(x^2+1)^2} \end{aligned}$$

We note that this formula remains valid for  $x = 0$ , as  $f$  vanishes on the whole  $y$ -axis.

Then we integrate in the  $x$ -direction,

$$\int_0^2 A(x) dx = \int_0^2 \frac{x dx}{2(x^2+1)^2} = \left[ \frac{-1}{4(x^2+1)} \right]_0^2 = -\frac{1}{20} + \frac{1}{4} = \frac{1}{5}$$

## Counterexample

**Second approach** We integrate first in the  $x$ -direction. For any  $y$ , we define  $B(y) = \int_0^2 f(x, y) dx$ . If  $y \neq 0$ , we again make the substitution  $u = x^2 + y^2$ , except now  $du = 2x dx$ . In view of the (skew-)symmetry of  $f$ , the integral is almost the same,

$$B(y) = \left[ -\frac{y}{2u} + \frac{y^3}{2u^2} \right]_{u=y^2}^{u=4+y^2} = -\frac{y}{2(4+y^2)} + \frac{y^3}{2(4+y^2)^2} = \frac{-2y}{(4+y^2)^2}$$

This formula too remains valid for  $y = 0$ , as  $f$  vanishes on the  $x$ -axis.

Then we integrate in the  $y$ -direction,

$$\int_0^1 B(y) dy = \int_0^1 \frac{-2y dy}{(4+y^2)^2} = \left[ \frac{1}{4+y^2} \right]_0^1 = \frac{1}{5} - \frac{1}{4} = -\frac{1}{20}$$

which disagrees with our first answer. Worse, this had to happen because  $B(y)$  is negative, while  $A(x)$  is positive.

## Counterexample

**Post-mortem** What happened here? The fact that all the functions we integrated are continuous functions of one variable offers no clue that anything is wrong. The point is that Fubini's Theorem *does not apply*, because the function  $f$  is not integrable over  $R$ ; indeed, it is not even bounded on  $R$ . (Nor is it Lebesgue-integrable.) It is continuous away from 0 but has a bad discontinuity at 0. What makes this counterexample work is that  $f$  takes arbitrarily large positive and negative values near the origin; explicitly,  $f(2t, t) = 6/125t^2$  and  $f(t, 2t) = -6/125t^2$  for any  $t \neq 0$ .



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# **Vector Calculus**

## **Fifth Edition**

### **Chapter 5:**

### **Double and Triple Integrals**

#### 5.3 The Double Integral Over More General Regions

## 5.3 The Double Integral Over More General Regions

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### Key Points in this Section.

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1. **Elementary Regions.** A *y-simple region* is one that lies between two continuous curves  $y = \phi_1(x)$  and  $y = \phi_2(x)$ , where  $\phi_1(x) \leq \phi_2(x)$  and  $a \leq x \leq b$ . Similarly, *x-simple regions* are those lying between two continuous curves  $x = \psi_1(y)$  and  $x = \psi_2(y)$ , where  $\psi_1(y) \leq \psi_2(y)$  and  $c \leq y \leq d$ . An *elementary region* is one that is either *y-simple* or is *x-simple*. If it is *both*, then it is called *simple*.
2. The integral of a function  $f$  over an elementary region  $D$  is obtained by extending  $f$  to  $f^*$ , the function defined to be  $f$  on  $D$  and zero outside  $D$  but inside a containing rectangle  $R$ . The integral of  $f$  over  $D$  is defined by

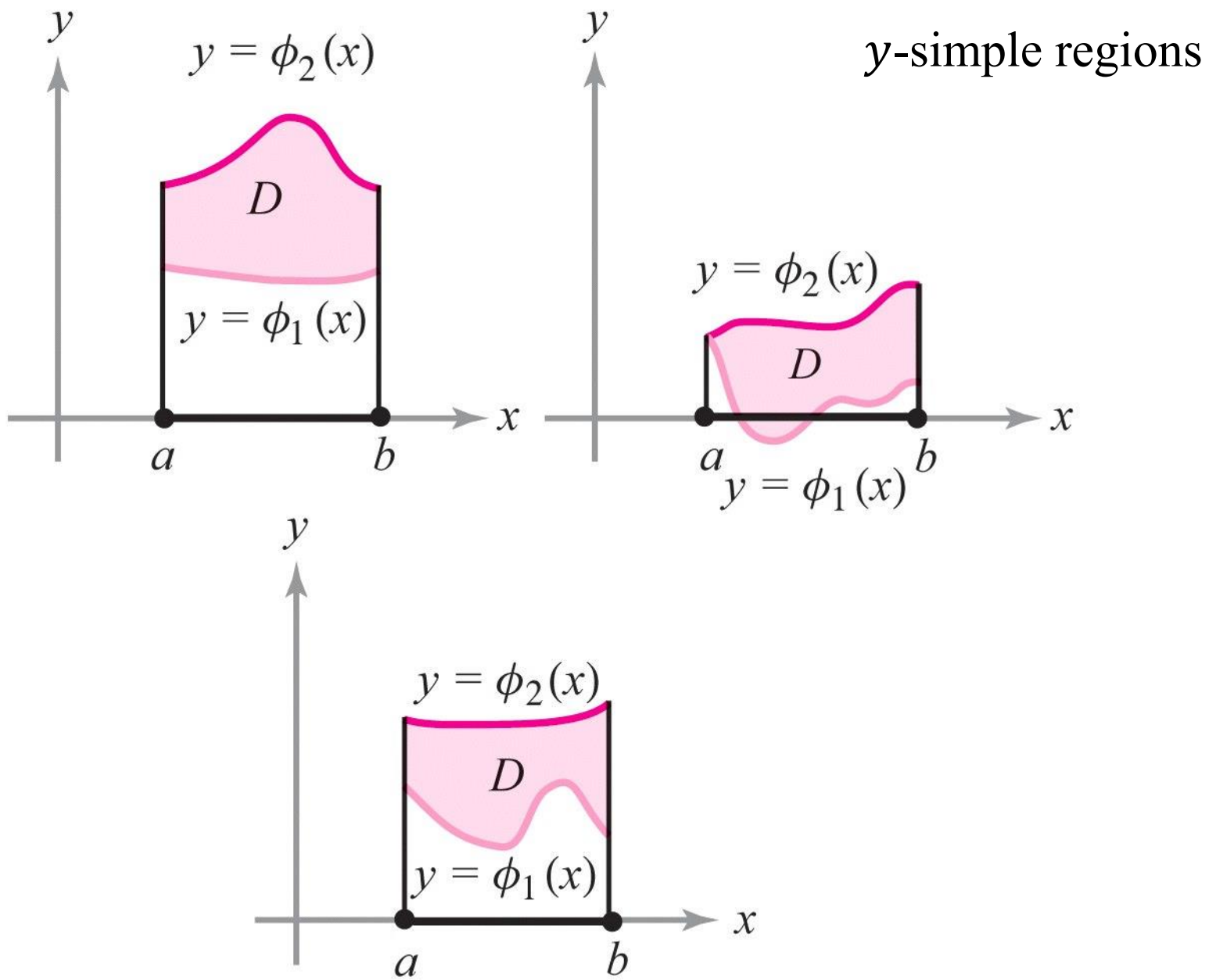
$$\iint_D f dA = \iint_R f^* dA.$$

3. For a  $y$ -simple region

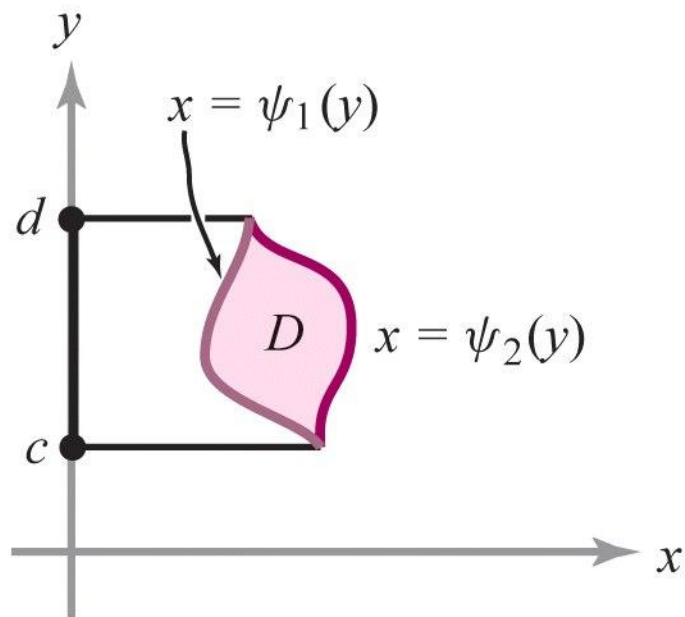
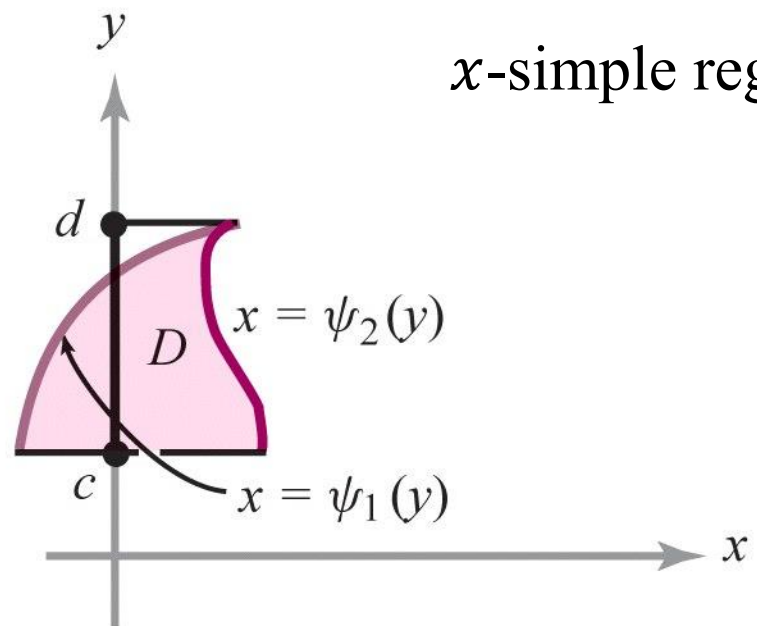
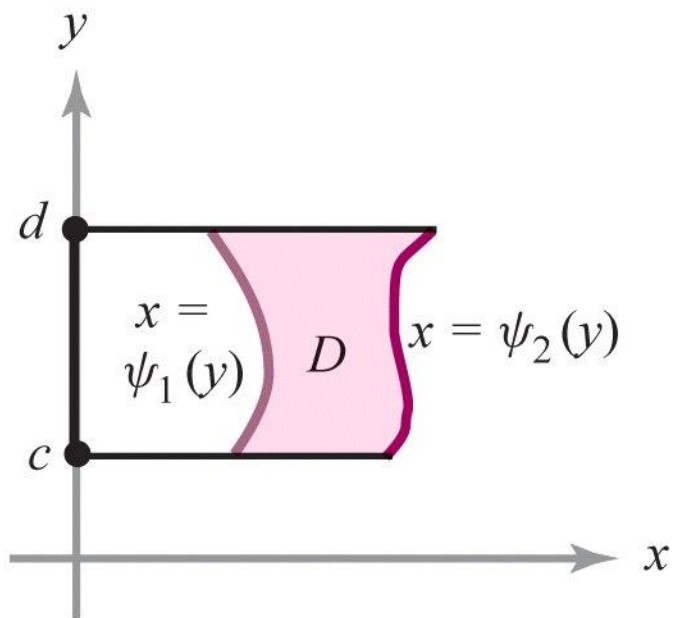
$$\iint_D f dA = \int_a^b \int_{\phi_1(x)}^{\phi_2(x)} f(x, y) dy dx$$

and for an  $x$ -simple region

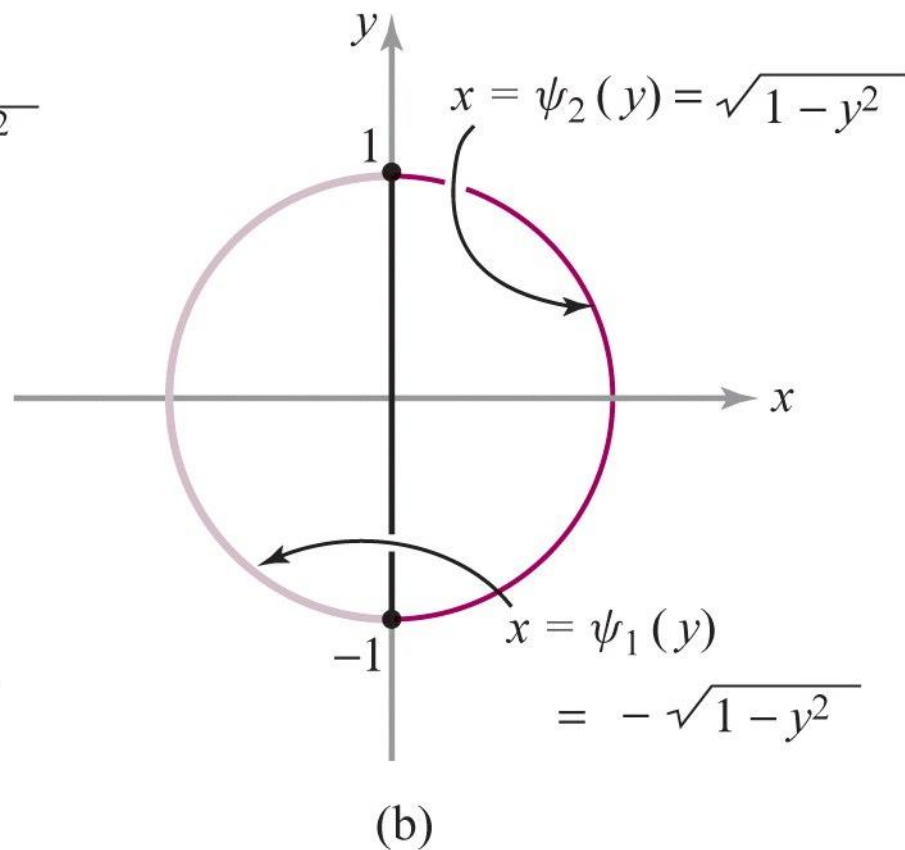
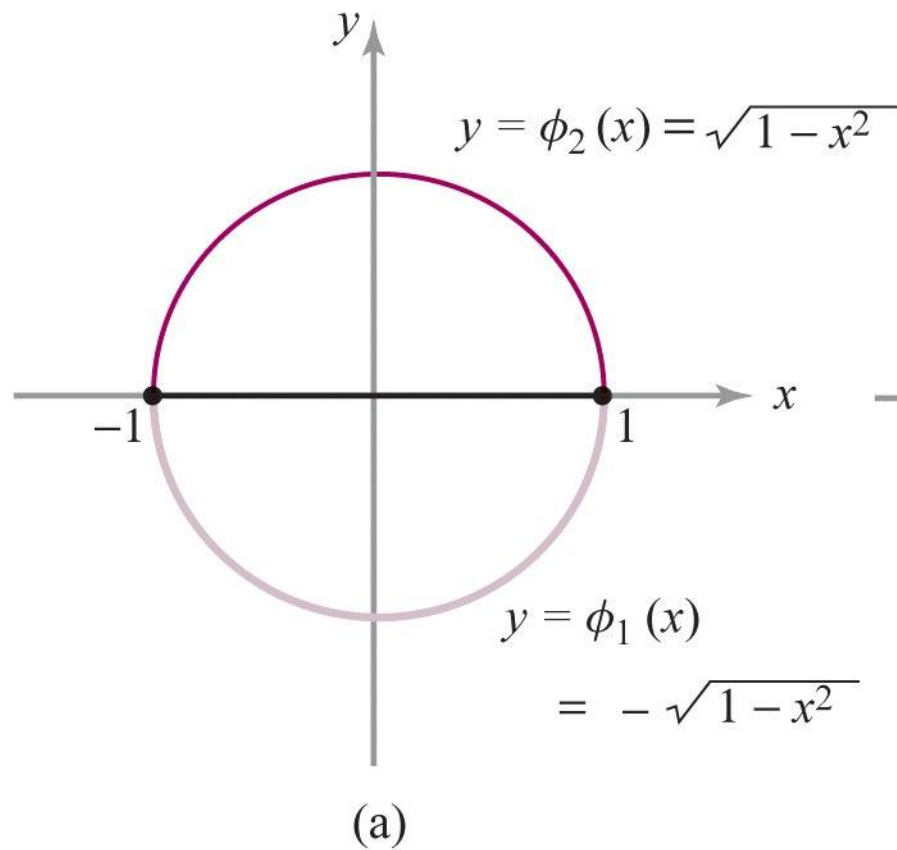
$$\iint_D f dA = \int_c^d \int_{\psi_1(y)}^{\psi_2(y)} f(x, y) dx dy.$$



$x$ -simple regions



# Elementary regions

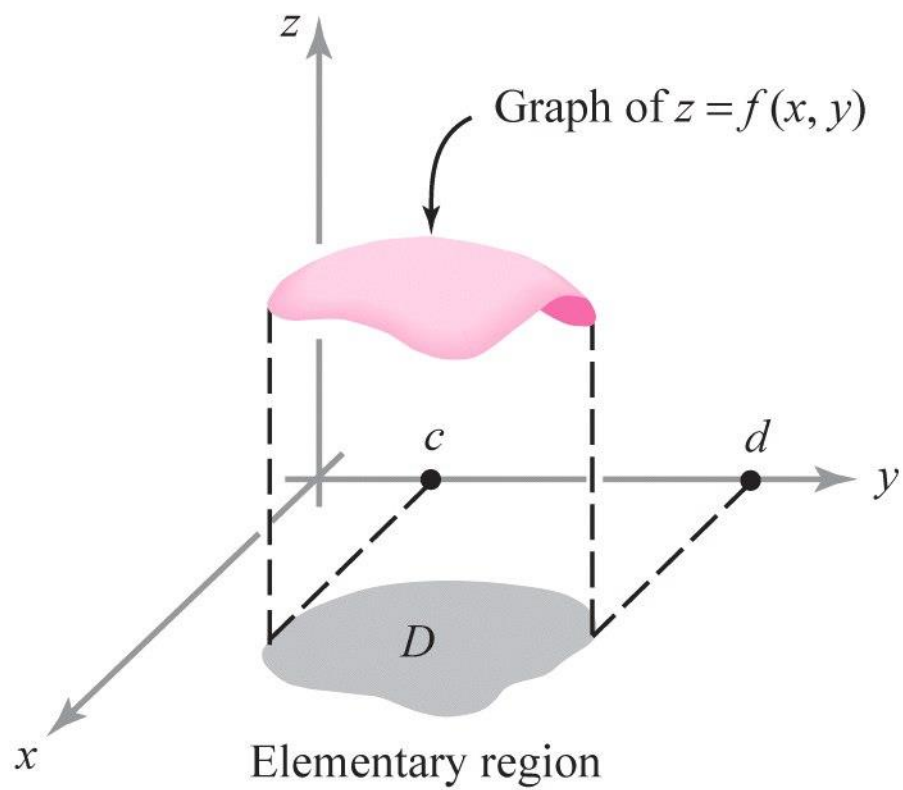


**DEFINITION: Integral over an Elementary Region** If  $D$  is an elementary region in the plane, choose a rectangle  $R$  that contains  $D$ . Given  $f: D \rightarrow \mathbb{R}$ , where  $f$  is continuous (and hence bounded), define  $\iint_D f(x, y) dA$ , the **integral of  $f$  over the set  $D$**  as follows: Extend  $f$  to a function  $f^*$  defined on all of  $R$  by

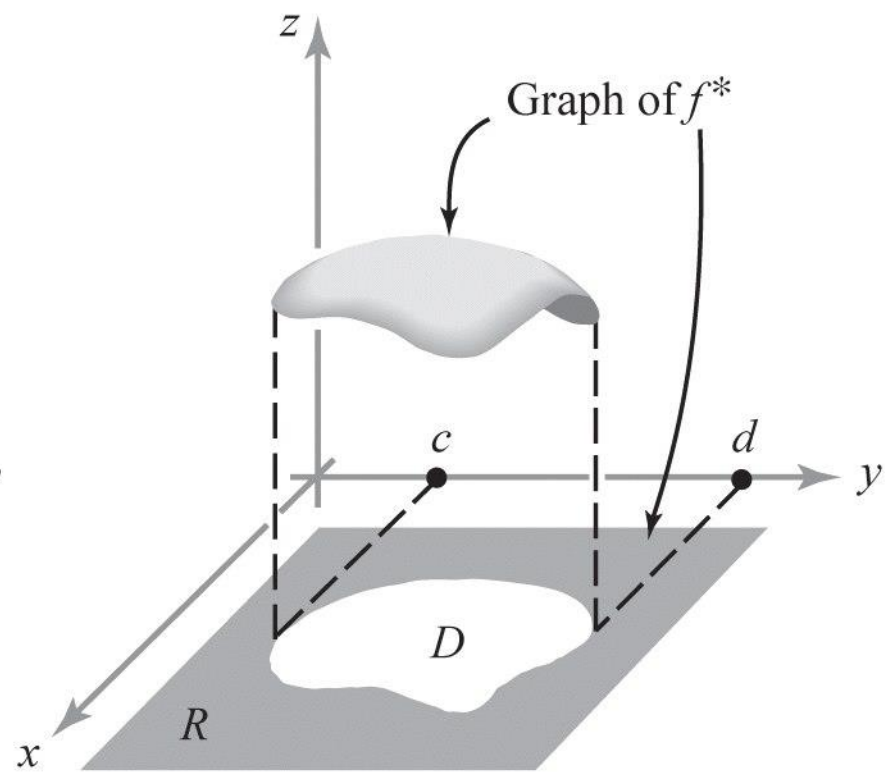
$$f^*(x, y) = \begin{cases} f(x, y) & \text{if } (x, y) \in D \\ 0 & \text{if } (x, y) \notin D \text{ and } (x, y) \in R. \end{cases}$$

Note that  $f^*$  is bounded (because  $f$  is) and continuous except possibly on the boundary of  $D$  (see Figure 5.3.4). The boundary of  $D$  consists of graphs of continuous functions, and so  $f^*$  is integrable over  $R$  by Theorem 2, Section 5.2. Therefore, we can define

$$\iint_D f(x, y) dA = \iint_R f^*(x, y) dA.$$



(a)

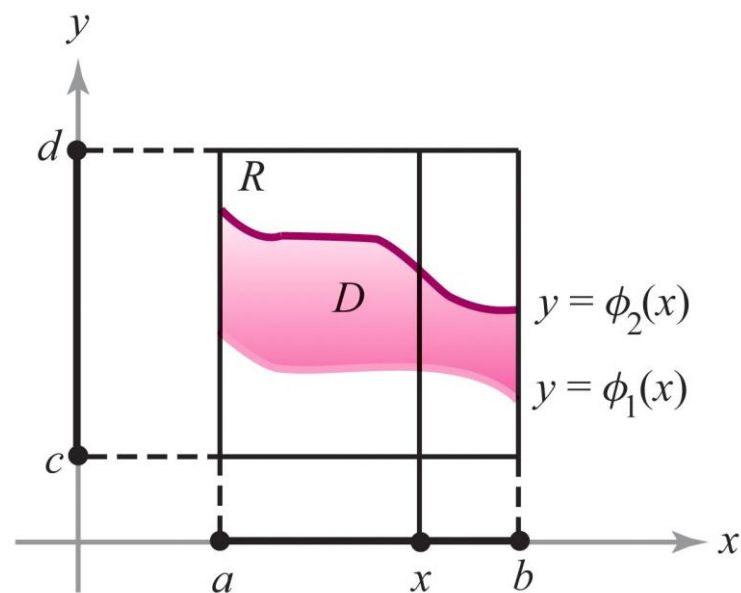


(b)



**THEOREM 4: Reduction to Iterated Integrals** If  $D$  is a  $y$ -simple region,

$$\iint_D f(x, y) dA = \int_a^b \int_{\phi_1(x)}^{\phi_2(x)} f(x, y) dy dx. \quad (1)$$



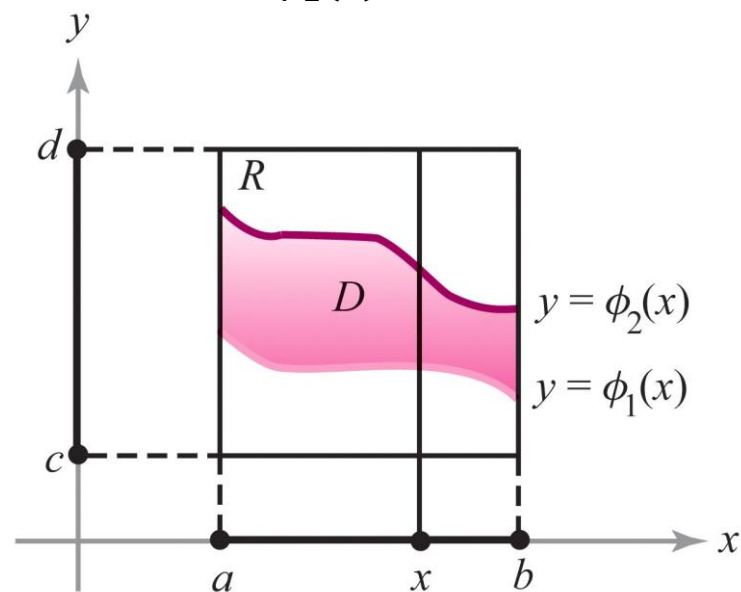
**THEOREM 4: Reduction to Iterated Integrals** If  $D$  is a  $y$ -simple region,

$$\iint_D f(x, y) dA = \int_a^b \int_{\phi_1(x)}^{\phi_2(x)} f(x, y) dy dx. \quad (1)$$

$$\int_c^d f^*(x, y) dy = \int_c^{\phi_1(x)} f^*(x, y) dy + \int_{\phi_1(x)}^{\phi_2(x)} f^*(x, y) dy + \int_{\phi_2(x)}^d f^*(x, y) dy$$

$$= 0 + \int_{\phi_1(x)}^{\phi_2(x)} f(x, y) dy + 0$$

$$= \int_{\phi_1(x)}^{\phi_2(x)} f(x, y) dy$$

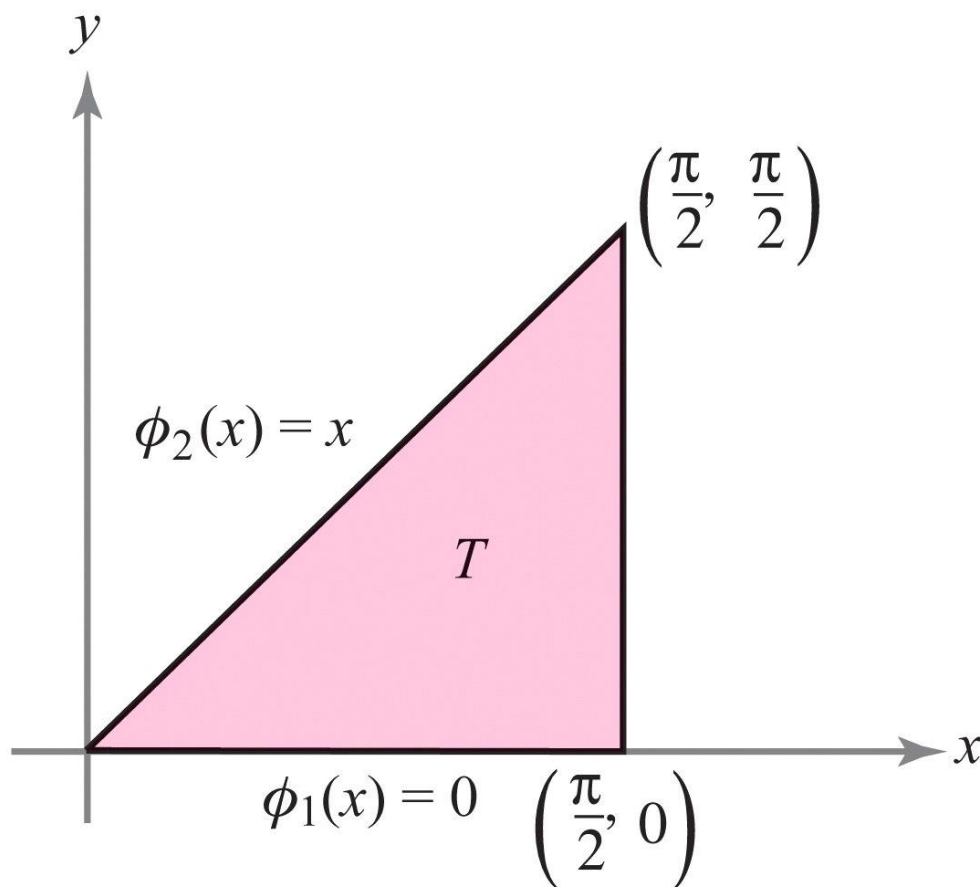


## Exercise

Find  $\iint_T (x^3 y + \cos x) \, dA$ , where  $T$  is the triangle consisting of all points  $(x, y)$  such that  $0 \leq x \leq \pi/2$ ,  $0 \leq y \leq x$ .

## Exercise

Find  $\iint_T (x^3 y + \cos x) dA$ , where  $T$  is the triangle consisting of all points  $(x, y)$  such that  $0 \leq x \leq \pi/2$ ,  $0 \leq y \leq x$ .



Solution:  $\frac{\pi^6}{768} + \frac{\pi}{2} - 1$

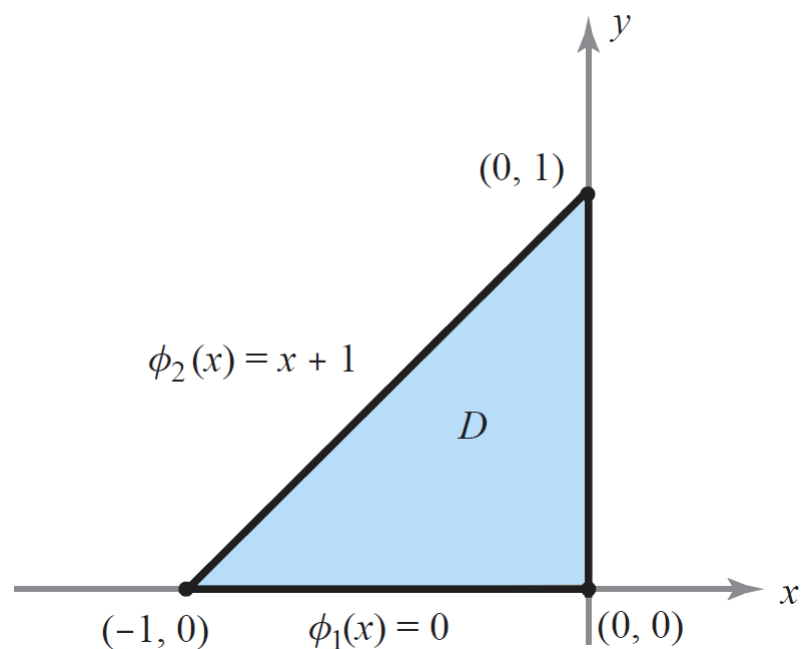
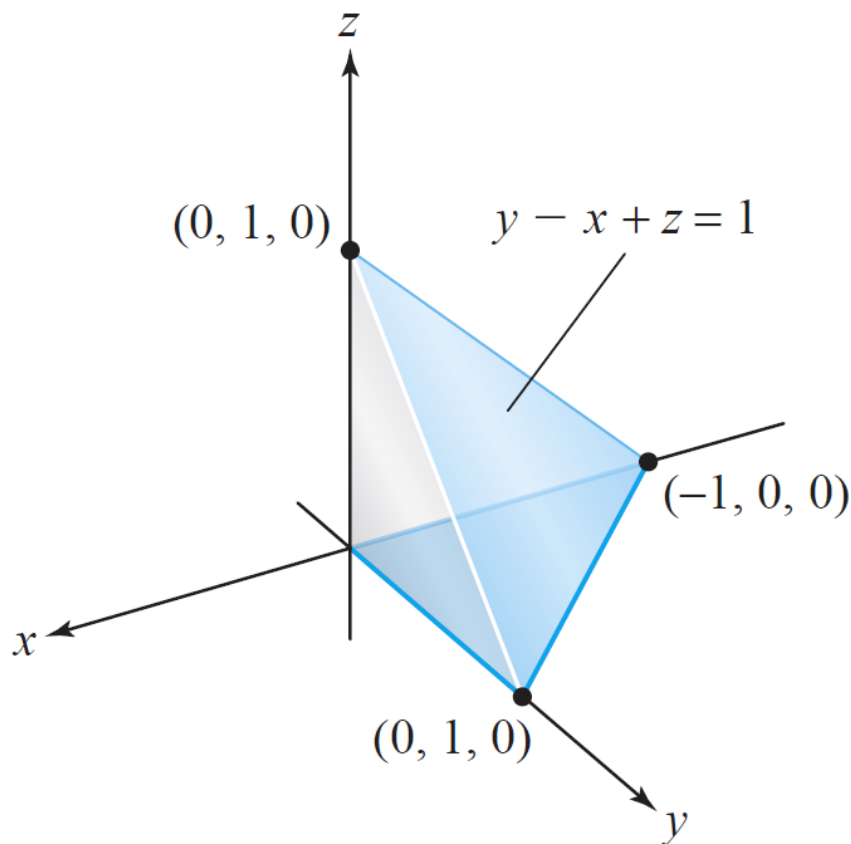
## Exercise

Find the volume of the tetrahedron bounded by the planes  $y=0$ ,  $z=0$ ,  $x=0$ , and  $y-x+z=1$

## Exercise

Find the volume of the tetrahedron bounded by the planes  $y=0$ ,  $z=0$ ,  $x=0$ , and  $y-x+z=1$

Solution:  $\frac{1}{6}$



**THEOREM 4': Iterated Integrals for  $x$ -Simple Regions** Suppose that  $D$  is the set of points  $(x, y)$  such that  $y \in [c, d]$  and  $\psi_1(y) \leq x \leq \psi_2(y)$ . If  $f$  is continuous on  $D$ , then

$$\iint_D f(x, y) dA = \int_c^d \left[ \int_{\psi_1(y)}^{\psi_2(y)} f(x, y) dx \right] dy. \quad (2)$$

Jerrold E. Marsden and Anthony J. Tromba

# **Vector Calculus**

## **Fifth Edition**

### **Chapter 5:**

### **Double and Triple Integrals**

#### 5.4 Changing the Order of Integration



## 5.4 Changing the Order of Integration

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### Key Points in this Section.

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1. If  $D$  is a *simple* region, that is, it is both  $x$ -simple and  $y$ -simple, then

$$\int_a^b \int_{\phi_1(x)}^{\phi_2(x)} f(x, y) dy dx = \int_c^d \int_{\psi_1(y)}^{\psi_2(y)} f(x, y) dx dy.$$

Sometimes one of these orders is simpler to evaluate than the other.

2. If  $m \leq f(x, y) \leq M$  on an elementary region  $D$ , then the *mean value inequality* holds:

$$m \text{ Area}(D) \leq \iint_D f dA \leq M \text{ Area}(D).$$

3. If  $f$  is continuous and  $D$  is an elementary region (that is, it is either  $x$ -simple or  $y$ -simple), then the *mean value equality* holds:

$$\iint_D f(x, y) dA = f(x_0, y_0) \text{ Area}(D).$$

for some point  $(x_0, y_0)$  in  $D$ .

## Exercise

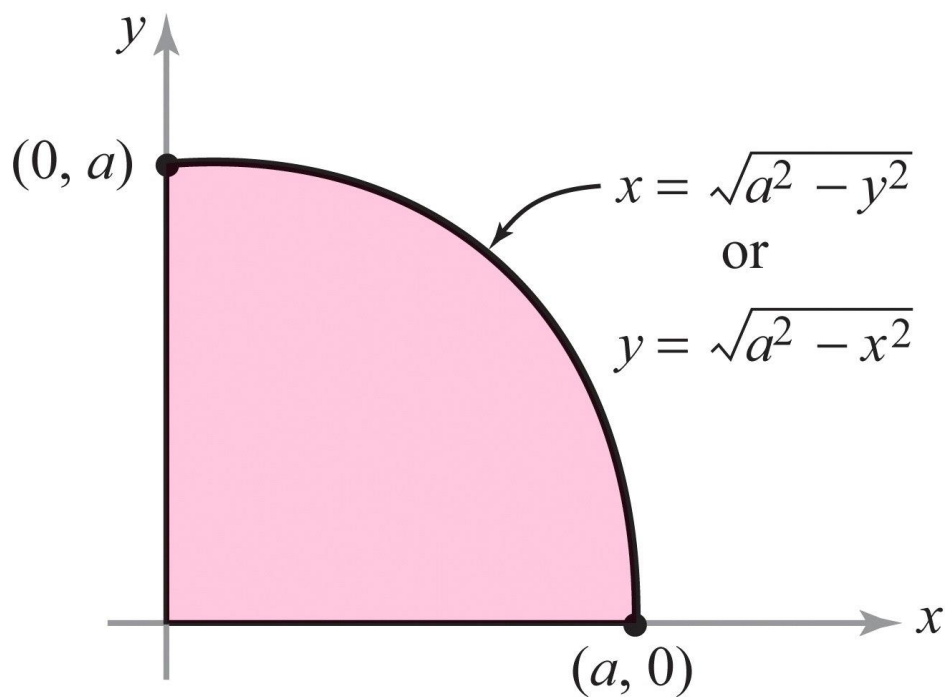
By changing the order of integration, evaluate

$$\int_0^a \int_0^{(a^2-x^2)^{1/2}} (a^2 - y^2)^{1/2} dy dx.$$

## Exercise

By changing the order of integration, evaluate

$$\int_0^a \int_0^{(a^2-x^2)^{1/2}} (a^2 - y^2)^{1/2} dy dx.$$



Solution:  $\frac{2a^3}{3}$

## Exercise

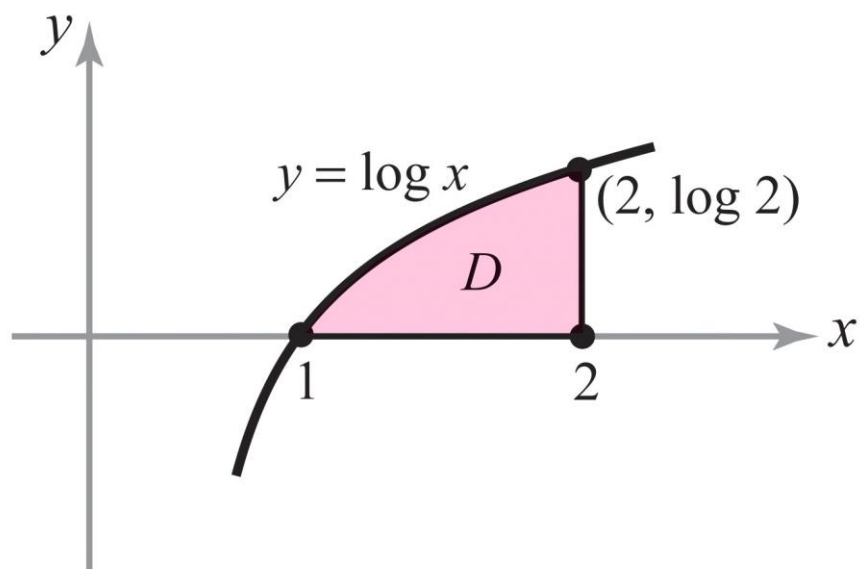
Evaluate

$$\int_1^2 \int_0^{\log x} (x-1) \sqrt{1+e^{2y}} \, dy \, dx.$$

## Exercise

Evaluate

$$\int_1^2 \int_0^{\log x} (x-1) \sqrt{1+e^{2y}} \, dy \, dx.$$



$$\text{Solution: } \frac{1}{2} \left( 2\sqrt{5} - \sqrt{2} + \log \frac{\sqrt{5} + 2}{\sqrt{2} + 1} \right) - \frac{1}{6} [5^{3/2} - 2^{3/2}]$$

**THEOREM 5: Mean Value Theorem: Double Integrals** Suppose  $f: D \rightarrow \mathbb{R}$  is continuous and  $D$  is an elementary region. Then for some point  $(x_0, y_0)$  in  $D$  we have

$$\iint_D f(x, y) dA = f(x_0, y_0)A(D),$$

where  $A(D)$  denotes the area of  $D$ .

Jerrold E. Marsden and Anthony J. Tromba

# **Vector Calculus**

## **Fifth Edition**

### **Chapter 5:**

### **Double and Triple Integrals**

#### 5.5 The Triple Integral

## 5.5 The Triple Integral

---

### Key Points in this Section.

---

1. **Definition of the Integral.** If  $f$  is a bounded function defined on a box  $B = [a, b] \times [c, d] \times [p, q]$  in  $\mathbb{R}^3$ , the triple integral, denoted

$$\iiint_B f dV, \quad \iiint_B f(x, y, z) dV, \quad \text{or} \quad \iiint_B f(x, y, z) dx dy dz$$

is defined as a limit of Riemann sums analogous to that for double integrals; if the limit exists,  $f$  is called *integrable*.

2. **Reduction to Iterated Integrals.** When  $f$  is integrable and an iterated integral exists, one has equality; for example,

$$\iiint_B f dV = \int_a^b \left\{ \int_p^q \left[ \int_c^d f(x, y, z) dy \right] dz \right\} dx$$



3. **Elementary Regions.** An example of an *elementary region*  $W$  in  $\mathbb{R}^3$  is one defined by inequalities  $a \leq x \leq b$ ,  $\phi_1(x) \leq y \leq \phi_2(x)$  (an elementary region in the plane) and  $\gamma_1(x, y) \leq z \leq \gamma_2(x, y)$ .
4. The *integral*  $\iiint_W f dV$  of a function  $f$  defined on an elementary region  $W$  is obtained, as for double integrals, by extending  $f$  to be zero outside  $W$  but inside a box  $B$  containing  $W$ .
5. For the elementary region  $W$  described in point 3,

$$\iiint_W f dV = \int_a^b \int_{\phi_1(x)}^{\phi_2(x)} \int_{\gamma_1(x,y)}^{\gamma_2(x,y)} f(x, y, z) dz dy dx.$$

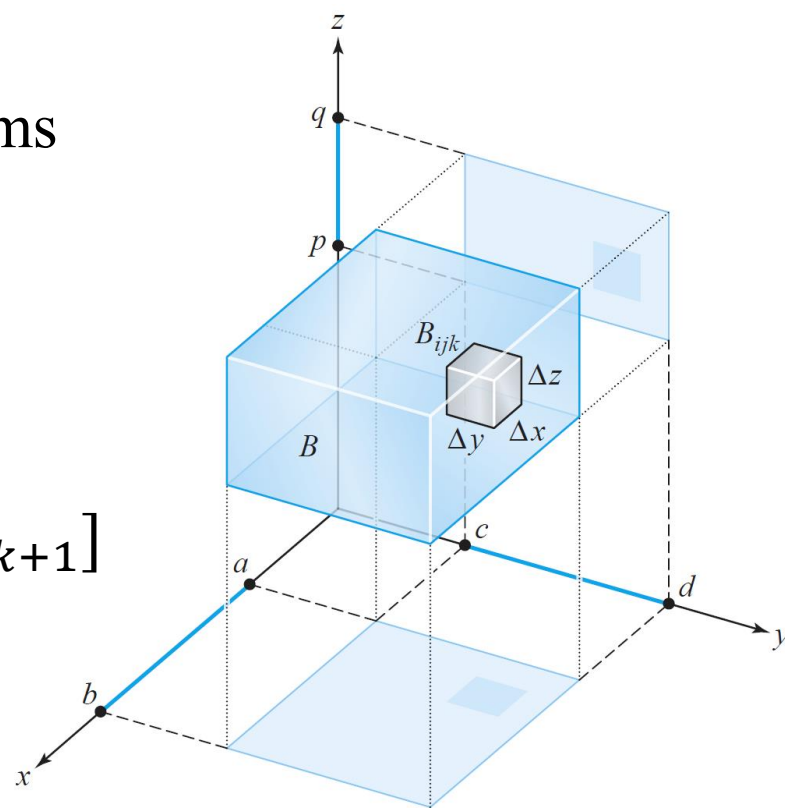
6. For regions that can be described as elementary regions in more than one way, one can, as with double integrals, change the order of integration.

## Triple integral as limit of Riemann sums

$$S_n = \sum_{i=0}^{n-1} \sum_{j=0}^{n-1} \sum_{k=0}^{n-1} f(\mathbf{c}_{ijk}) \Delta V$$

$$\mathbf{c}_{ijk} \in [x_i, x_{i+1}] \times [y_j, y_{j+1}] \times [z_k, z_{k+1}]$$

$$\Delta V = \Delta x \Delta y \Delta z$$



**DEFINITION: Triple Integrals** Let  $f$  be a bounded function of three variables defined on  $B$ . If  $\lim_{n \rightarrow \infty} S_n = S$  exists and is independent of any choice of  $\mathbf{c}_{ijk}$ , we call  $f$  **integrable** and call  $S$  the **triple integral** (or simply the integral) of  $f$  over  $B$  and denote it by

$$\iiint_B f \, dV, \quad \iiint_B f(x, y, z) \, dV \quad \text{or} \quad \iiint_B f(x, y, z) \, dx \, dy \, dz.$$

**Reduction to Iterated Integrals** Let  $f(x, y, z)$  be integrable on the box  $B = [a, b] \times [c, d] \times [p, q]$ . Then any iterated integral that exists is equal to the triple integral; that is,

$$\begin{aligned}\iiint_B f(x, y, z) \, dx \, dy \, dz &= \int_p^q \int_c^d \int_a^b f(x, y, z) \, dx \, dy \, dz \\ &= \int_p^q \int_a^b \int_c^d f(x, y, z) \, dy \, dx \, dz \\ &= \int_a^b \int_p^q \int_c^d f(x, y, z) \, dy \, dz \, dx,\end{aligned}$$

and so on. (There are six possible orders altogether.)

## Exercise

- (a) Let  $B$  be the box  $[0, 1] \times [-\frac{1}{2}, 0] \times [0, \frac{1}{3}]$ . Evaluate

$$\iiint_B (x + 2y + 3z)^2 dx dy dz.$$

- (b) Verify that we get the same answer if the integration is done in the order  $y$  first, then  $z$ , and then  $x$ .

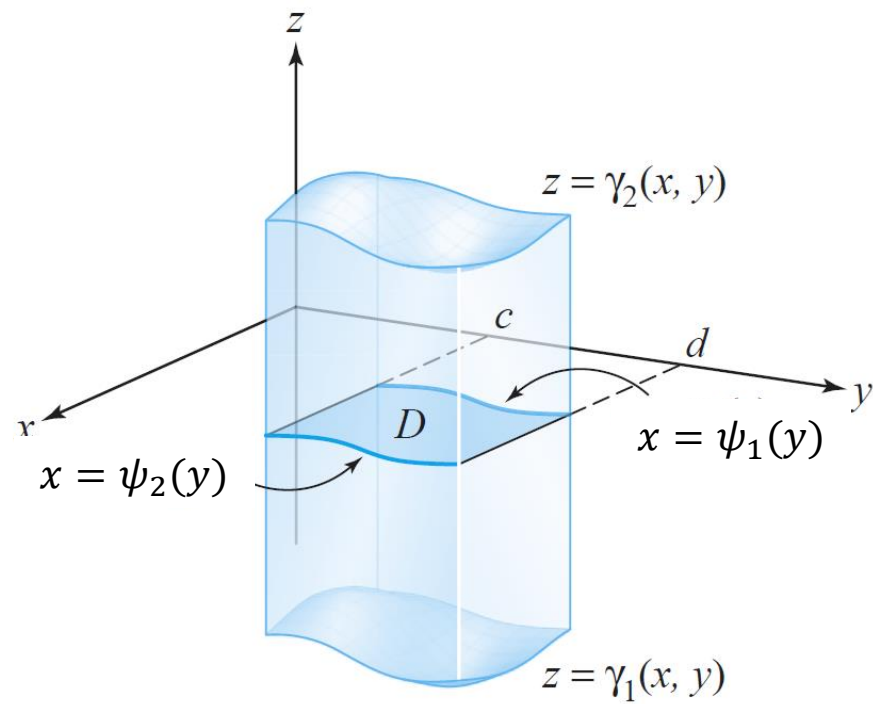
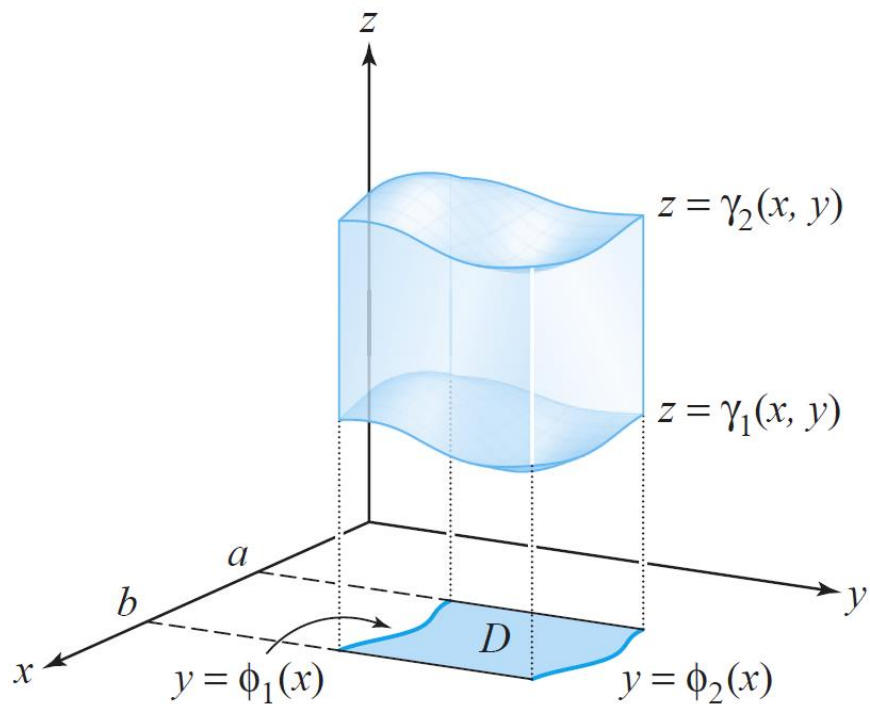
Solution:  $\frac{1}{12}$

## Exercise

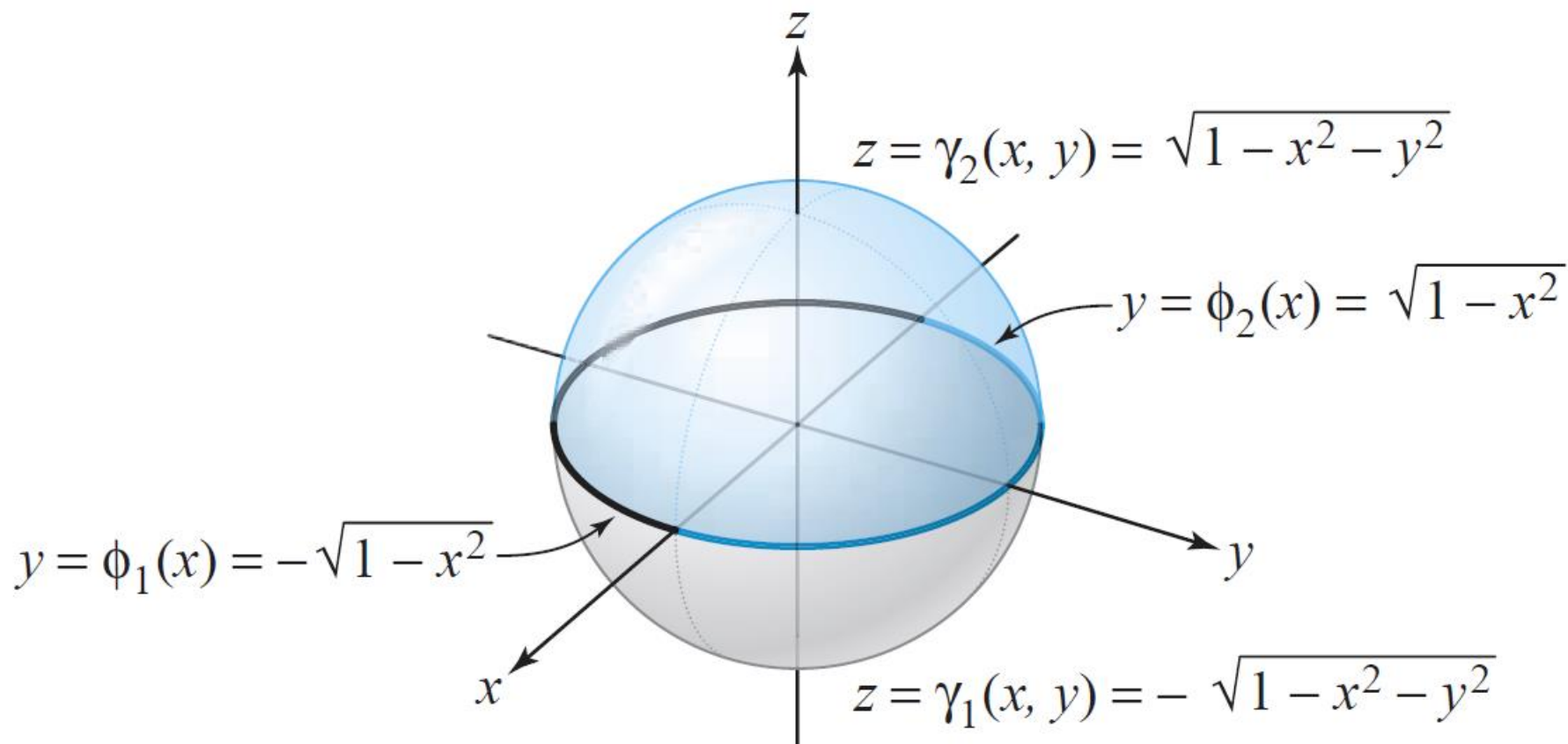
Integrate  $e^{x+y+z}$  over the box  $[0, 1] \times [0, 1] \times [0, 1]$

Solution:  $(e - 1)^3$

# Elementary regions



## Elementary regions



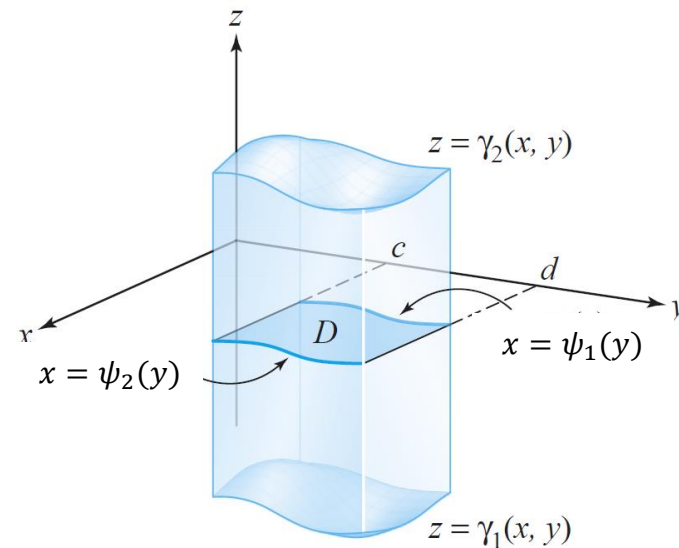
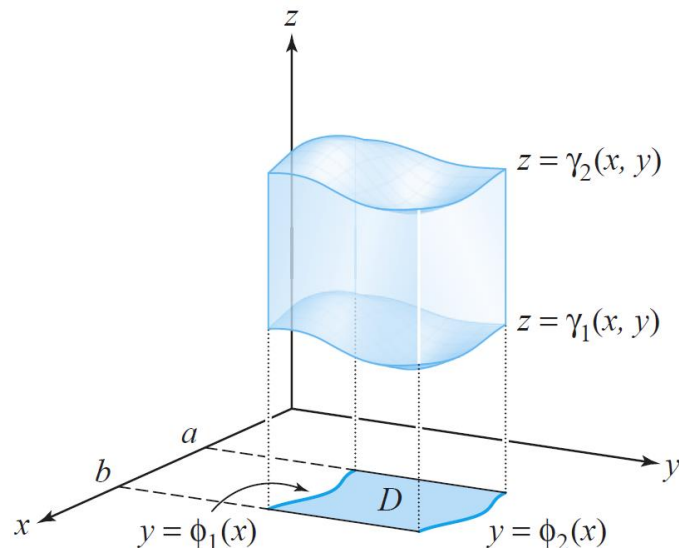
**Triple Integrals by Iterated Integration** Suppose that  $W$  is an elementary region described by bounding  $z$  between two functions of  $x$  and  $y$ . Then either

$$\iiint_W f(x, y, z) \, dx \, dy \, dz = \int_a^b \int_{\phi_1(x)}^{\phi_2(x)} \int_{\gamma_1(x, y)}^{\gamma_2(x, y)} f(x, y, z) \, dz \, dy \, dx$$

[see Figure 5.5.2 (left)] or

$$\iiint_W f(x, y, z) \, dx \, dy \, dz = \int_c^d \int_{\psi_1(y)}^{\psi_2(y)} \int_{\gamma_1(x, y)}^{\gamma_2(x, y)} f(x, y, z) \, dz \, dx \, dy$$

[see Figure 5.5.2 (right)].





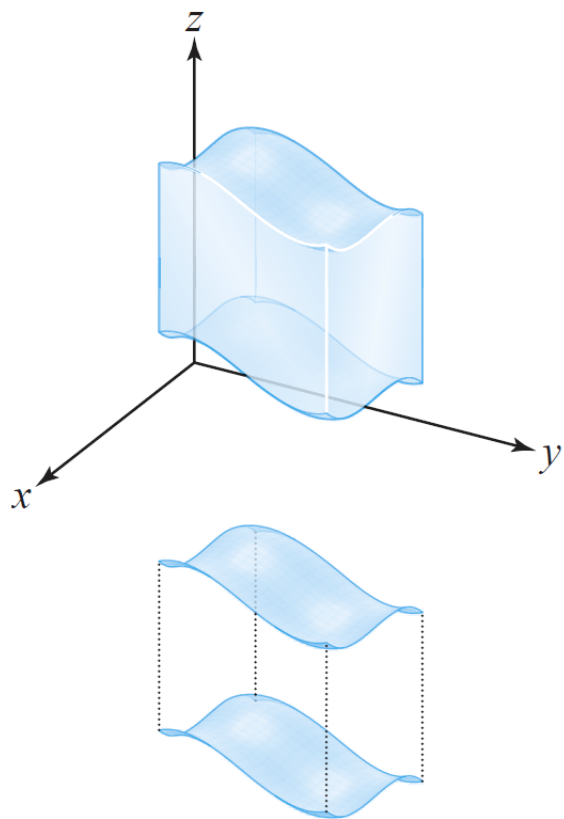
## Exercise

Verify that the volume formula for the ball of radius 1:

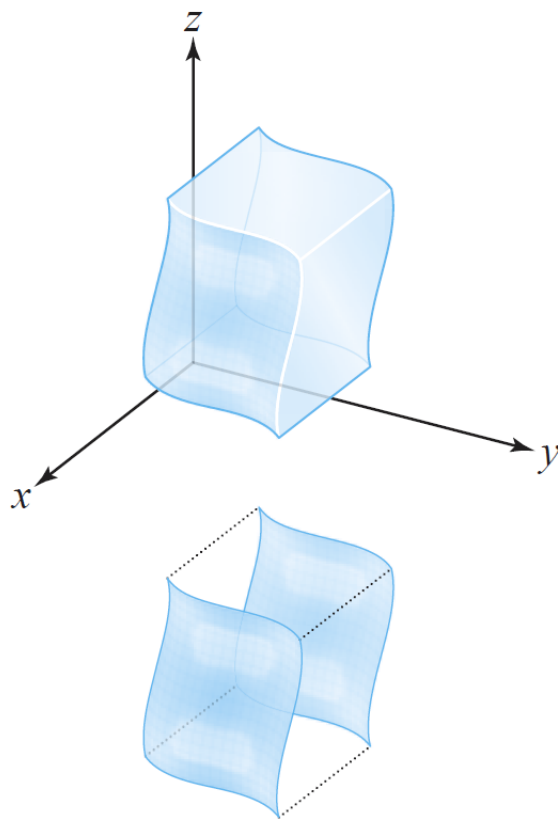
$$\iiint_W dx \, dy \, dz = \frac{4}{3}\pi,$$

where  $W$  is the set of  $(x, y, z)$  with  $x^2 + y^2 + z^2 \leq 1$ .

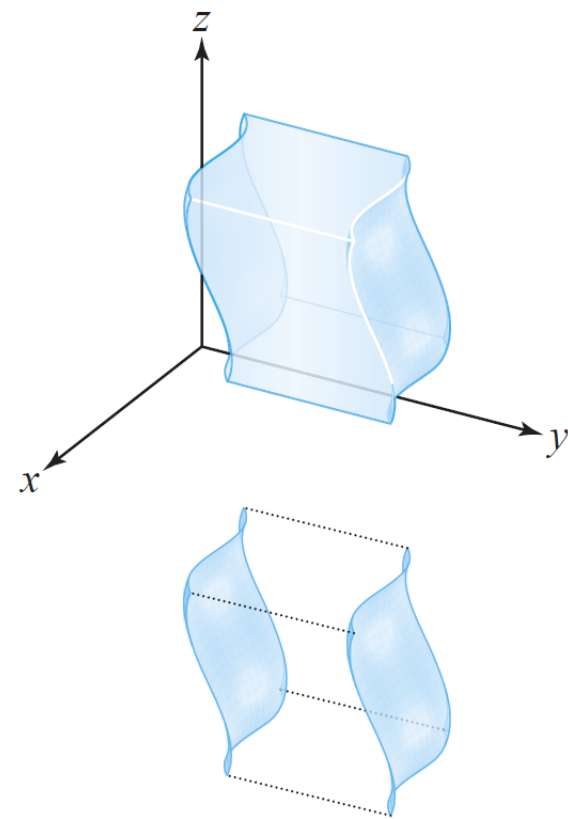
# Elementary regions



Top and bottom are  
surfaces  $z = \gamma(x, y)$



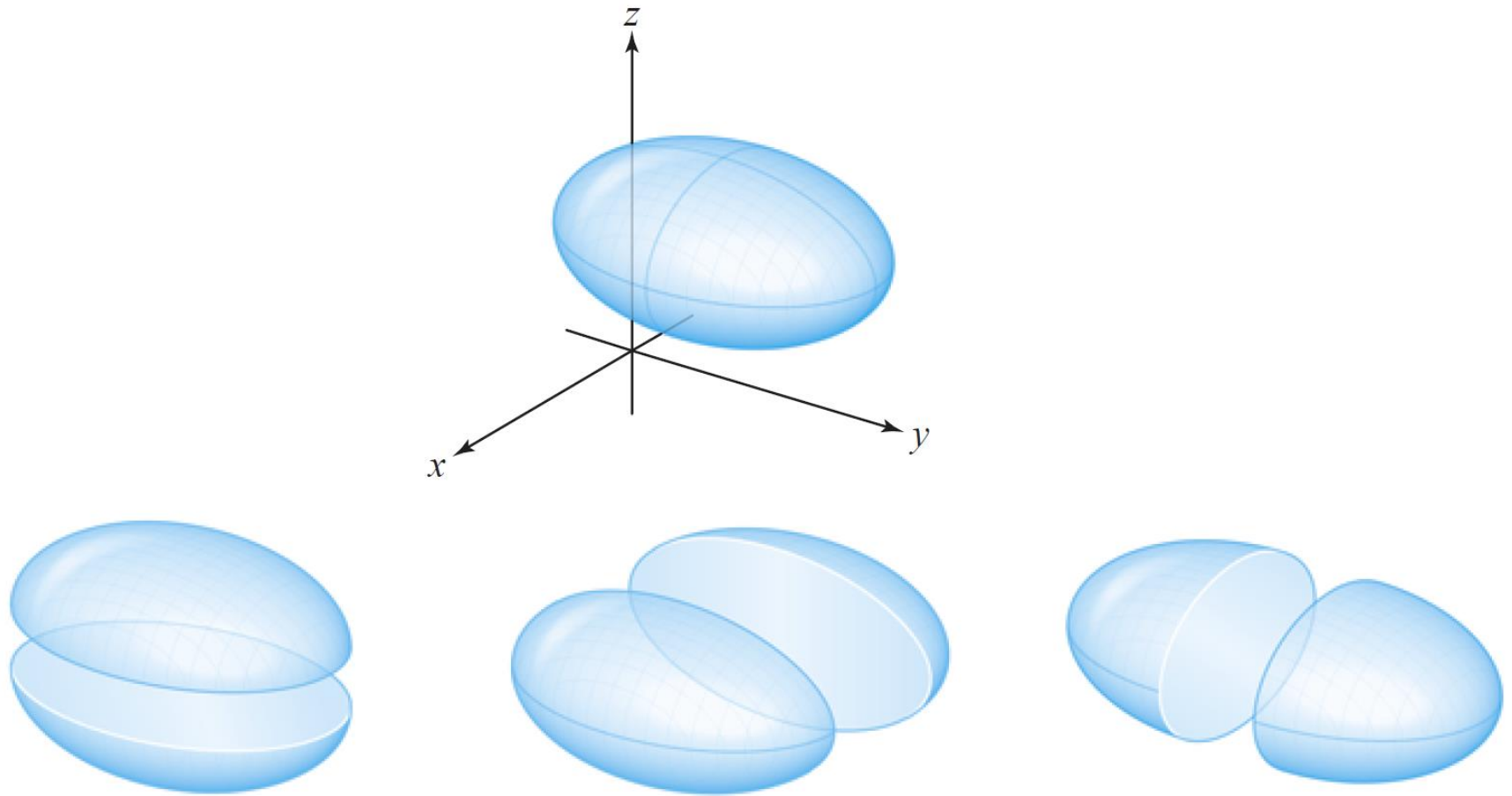
Front and rear are  
surfaces  $x = \rho(y, z)$



Left and side are  
surfaces  $y = \delta(x, z)$

# Symmetric elementary regions

They are elementary regions simultaneously for the three axes

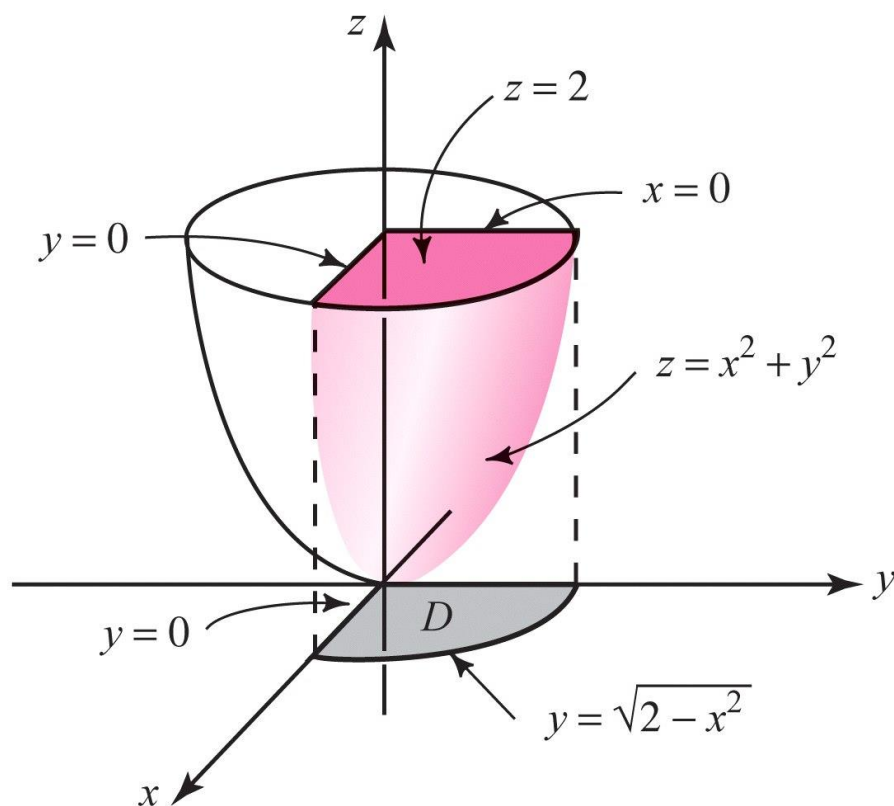


## Exercise

Let  $W$  be the region bounded by the planes  $x = 0$ ,  $y = 0$ , and  $z = 2$ , and the surface  $z = x^2 + y^2$  and lying in the quadrant  $x \geq 0$ ,  $y \geq 0$ . Compute  $\iiint_W x \, dx \, dy \, dz$  and sketch the region.

## Exercise

Let  $W$  be the region bounded by the planes  $x = 0$ ,  $y = 0$ , and  $z = 2$ , and the surface  $z = x^2 + y^2$  and lying in the quadrant  $x \geq 0$ ,  $y \geq 0$ . Compute  $\iiint_W x \, dx \, dy \, dz$  and sketch the region.



Solution:  $\frac{8\sqrt{2}}{15}$